

GCP Professional Cloud Architect Crash Course

Get Fully Prepared to Crush the Exam



Victor Dantas

Author, Certified Cloud Architect

Day 2 Schedule

- **Segment 5: Configuring network, storage, and compute**

- Configuring and securing networks
- Setting up hybrid and multicloud networking
- Provisioning and configuring data storage
- Configuring data security and access management
- Provisioning and configuring compute systems
- Configuring Kubernetes
- Building ML workflows and using Vertex AI APIs
- Sample exam questions

- **Segment 6: Designing for security and compliance**

- Identity and Access Management (IAM)
- Resource hierarchy
- Service controls
- Securing data at rest and in transit
- Secrets and certificate management
- Compliance considerations
- Securing AI
- Sample exam questions

Day 2 Schedule

- **Segment 7: Deployment and operational best practices**

- Google Cloud Observability (monitoring and logging, profiling, alerting)
- Troubleshooting and incident management
- Interacting with GCP programmatically
- Automating infrastructure and application deployments
- Testing and validating deployments
- API management best practices
- Gemini Cloud Assist
- Sample exam questions

- **Segment 8: Case studies solutioning**

- Case Study Breakdown: Altostrat Media
- Case Study Breakdown: Cymbal Retail
- Case Study Breakdown: EHR Healthcare
- Case Study Breakdown: KnightMotives Automotive

Segment 5: Configuring network, storage, and compute



Objectives

- Configuring and securing networks
- Setting up hybrid and multicloud networking
- Provisioning and configuring data storage
- Configuring data security and access management
- Provisioning and configuring compute systems
- Configuring Kubernetes
- Building ML workflows and using Vertex AI APIs
- Sample exam questions

Configuring VPC Networks

VPCs are **global** resources; subnets are **regional**. Each project starts with a default auto-mode VPC. **Custom mode is recommended for production.**

← Create a VPC network

Name *
my-vpc-network ⓘ
Lowercase letters, numbers, hyphens allowed

Description ⓘ

☒ Set MTU automatically ⓘ

Maximum transmission unit (MTU)
Auto-configured MTU ⓘ

☐ Configure network profile ⓘ

Subnet creation mode ⓘ

☒ Custom

☐ Automatic

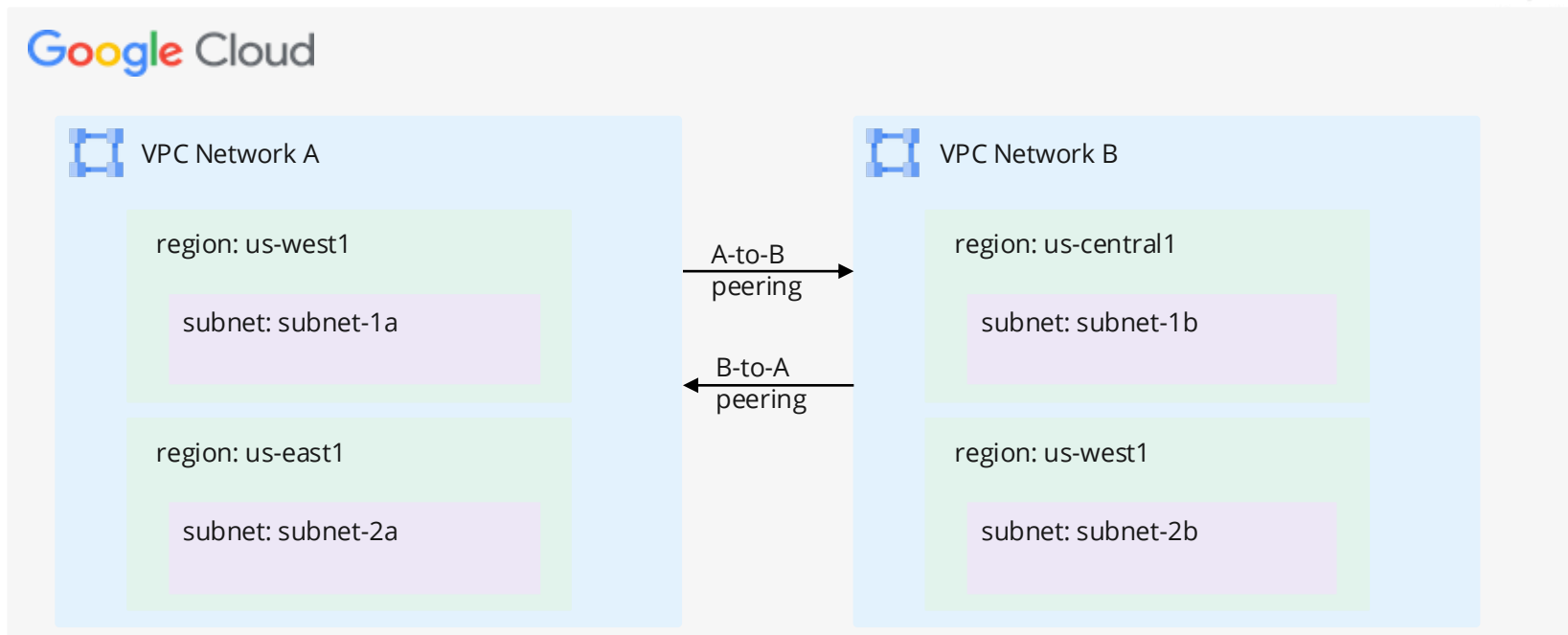
Key Exam Concepts

- **Secondary ranges** : alias IPs for GKE pods/services
- **Flow Logs** : VPC flow logging for network monitoring

Subnet Stack Types

- **IPv4-only** — default for most workloads
- **Dual-stack** — IPv4 + IPv6 (custom mode only)
- **IPv6-only** — emerging use cases

Configuring VPC Peering



Exam Tip: Requires non-overlapping ranges. Two auto-mode VPCs cannot peer (identical ranges).

Configuring Private Service Connect

PSC uses a **consumer-producer** model. The consumer creates an endpoint in their VPC; the producer publishes a service attachment. NAT between sides eliminates IP overlap concerns, so no VPC peering required.

1. Access Google APIs

Create a PSC endpoint with a private IP you choose. Configure DNS to resolve ***.googleapis.com** to that IP. All Google APIs accessible via a single controlled endpoint. Cleaner than PGA shared VIP ranges.

2. Access Managed Services

Google services (Cloud SQL, AlloyDB, Memorystore) publish a **service attachment**. You create an endpoint connecting to it. Service appears at a private IP you control — no allocated ranges or VPC peering needed.

3. Publish Your Own Services

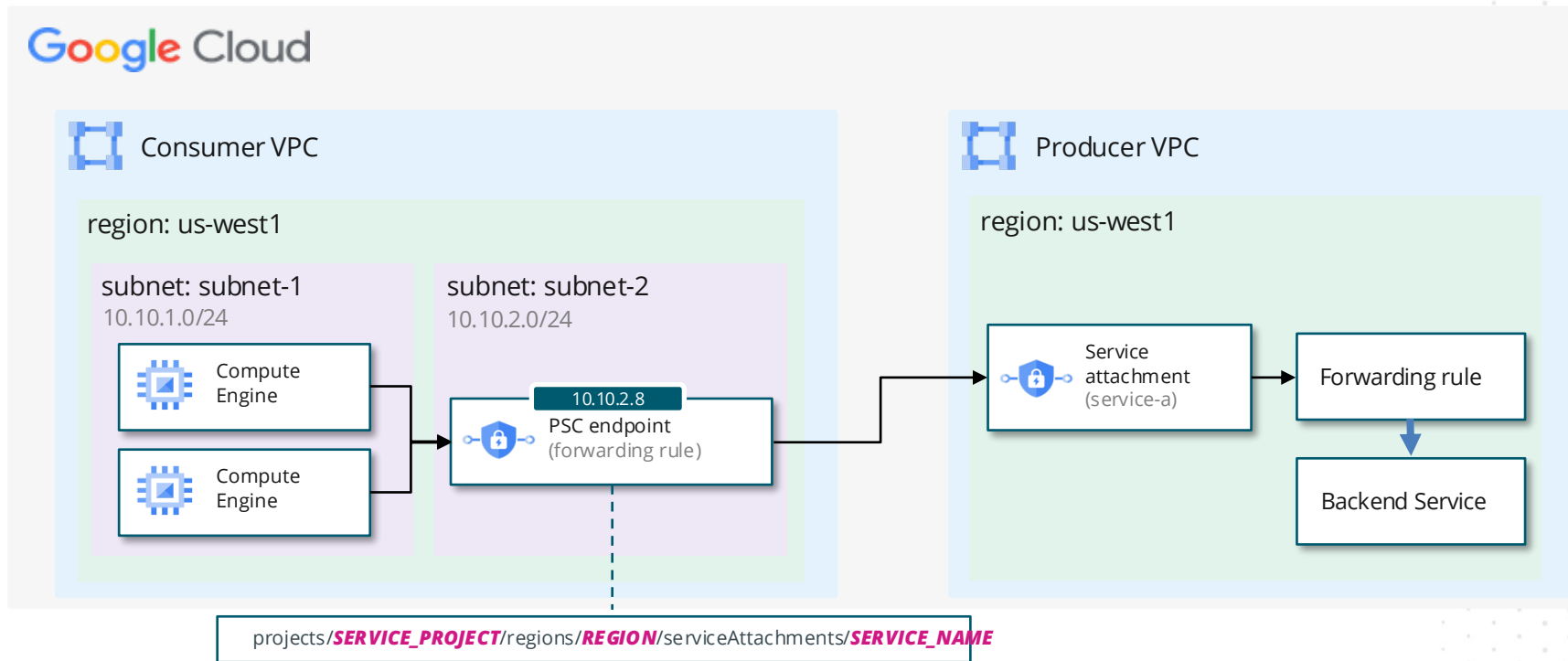
Deploy an Internal Load Balancer → create a service attachment exposing it → other VPCs or orgs create endpoints to consume. Requires a dedicated **NAT subnet** for source address translation on the producer side.

Consumer side: Endpoint (forwarding rule) → private IP in your VPC → routes to service attachment

Producer side: Service attachment → NAT subnet → Internal Load Balancer → backends

Exam Tip: PSC is Google's recommended approach for new private connectivity.

Configuring Private Service Connect



Firewall Rules and Policies

Google Cloud evaluates firewall rules in a specific order across four layers. Higher layers take precedence. Each layer can **allow**, **deny**, or **delegate** (goto_next) to the next layer.

1. Hierarchical Firewall Policies (Org/Folder)

Highest precedence. Org-wide enforcement. Targets VPCs or service accounts. Use for blocking known-bad IPs, allowing health checks.

2. VPC Firewall Rules (Classic)

Per-VPC, individual rules. Target via network tags or service accounts. Default: evaluated before network firewall policies.

3. Network Firewall Policies (Global & Regional)

Batch-managed as a policy object. Support IAM-governed Tags (more secure than network tags). Evaluation order configurable: BEFORE or AFTER classic rules.

4. Implied Rules

Default deny all ingress, default allow all egress. Cannot be deleted but can be overridden by higher-priority rules.

Default Evaluation Order: Hierarchical → VPC Firewall Rules → Global Network Policy → Regional Network Policy → Implied Rules

Firewall Targets: Tags vs Service Accounts

Firewall rules use **targets** to determine which instances a rule applies to. The three target mechanisms differ in security, scope, and management model.

Source Filtering

Ingress: Source IP ranges, source tags, or source service accounts. Tags and SAs cannot be combined in one rule.

Egress: Destination IP ranges only (no tag/SA filtering on egress destinations).

Target Types in Rules

All instances — rule applies to every VM in the VPC.

By tag — applies only to VMs with the specified network tag.

By service account — applies to VMs running as that SA.

Best Practice

Use **service accounts** for firewall targeting in production. Identity is IAM-bound and cannot be added by any instance editor.

The screenshot shows the configuration for a Firewall rule. The 'Direction of traffic' is set to 'Ingress'. The 'Action on match' is set to 'Allow'. The 'Targets' section is set to 'Specified target tags', with 'web-server' selected. The 'Source filter' is set to 'IPv4 ranges', with '0.0.0.0/24' selected. The 'Second source filter' is set to 'None'. The 'Destination filter' is set to 'None'. The 'Protocols and ports' section is expanded, showing 'TCP' selected with port '443'. The 'UDP' and 'SCTP' options are unchecked. The 'Other' option is also unchecked. The 'Protocols' field is empty. A note at the bottom states: 'Separate multiple protocols by commas, e.g. AH, ICMP'.

NGFW Tiers

NGFW Essentials

Firewall policies

Address groups

Network tags and service account filters

NGFW Standard

Essentials plus...

FQDN objects

Geolocation objects

Google Threat Intelligence

NGFW Enterprise

Standard plus...

Advanced layer 7 security

Cloud Armor and Load Balancing

Cloud Armor

DDoS protection + WAF for apps behind load balancers. L3/L4 DDoS is automatic. L7 WAF rules for OWASP Top 10.

- **Adaptive Protection** — ML-based L7 DDoS detection
- **Rate limiting** — throttle per client/path
- **Bot management** — reCAPTCHA Enterprise
- **IP/Geo blocking** — by range, country, ASN

Cloud Armor has a custom rules language

Example expression: `inIpRange(origin.ip, '9.9.9.0/24')`

Load Balancer Types

- HTTP routing → Application LB
- TCP (non-HTTP) → Proxy Network LB
- Preserve client IP → Passthrough NLB
- UDP/ESP/ICMP → Passthrough NLB (only option)
- Cloud Armor → Global external ALB or proxy NLB

Exam Tip: Global LBs require Premium Tier networking. Cloud Armor only works with proxy-type LBs

Configuring VPC Service Controls

VPC-SC perimeters block API access across the boundary, **even with valid IAM permissions**.

Step 1: Create an Access Policy

One access policy per organization. Contains all perimeters and access levels. Requires `accesscontextmanager.policy.create` at org level.

Step 2: Define Access Levels

Context conditions for allowing access from outside. Combine with AND/OR: **IP ranges** (public CIDRs only), **device trust** (Endpoint Verification), **identity** (users/groups), **geography**.

Step 3: Create Service Perimeter

Add projects and restrict services (BigQuery, GCS, Cloud SQL, etc.). Start with **dry run mode** — logs violations to Cloud Logging without blocking. Convert to enforced mode once validated.

Step 4: Write Ingress/Egress Rules

Ingress: Allow specific external identities/IPs to call specific services inside.

Egress: Allow perimeter resources to access specific external projects/services.

DNS Integration

For full enforcement, configure VMs to use `restricted.googleapis.com` (199.36.153.4/30). This ensures API calls from within the perimeter are subject to VPC-SC checks.

Ingress/Egress Rule Anatomy

From: source identity + source (access level, project, VPC). **To:** target project + service + method. Rules are directional and can specify individual API methods for fine-grained control.

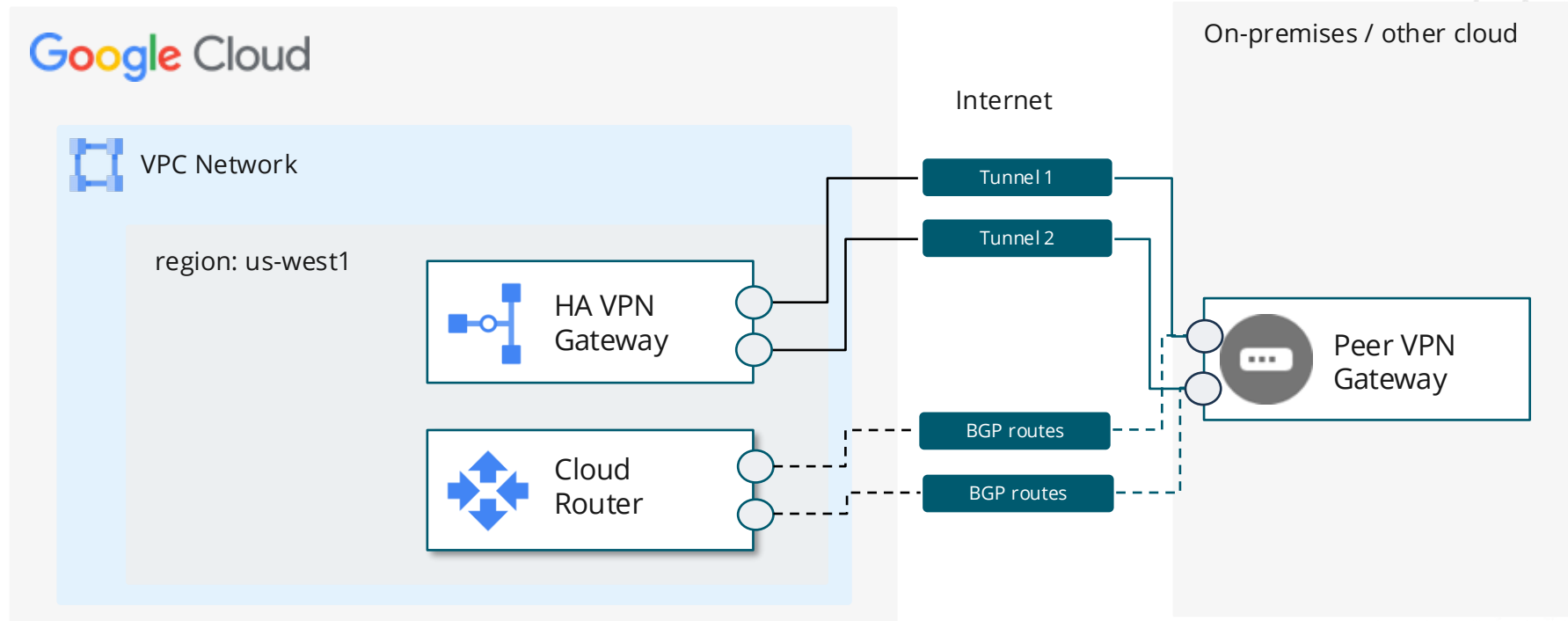
Example:

- Ingress rule: "Service account **ci-bot@partner.iam** from access level **corporate-vpn** can call **bigquery.datasets.get** on project **analytics-prod**"
- Egress rule: "**Cloud Function** in this perimeter can write to **GCS bucket** in project **external-data**"

Securing Networks: General Guidance

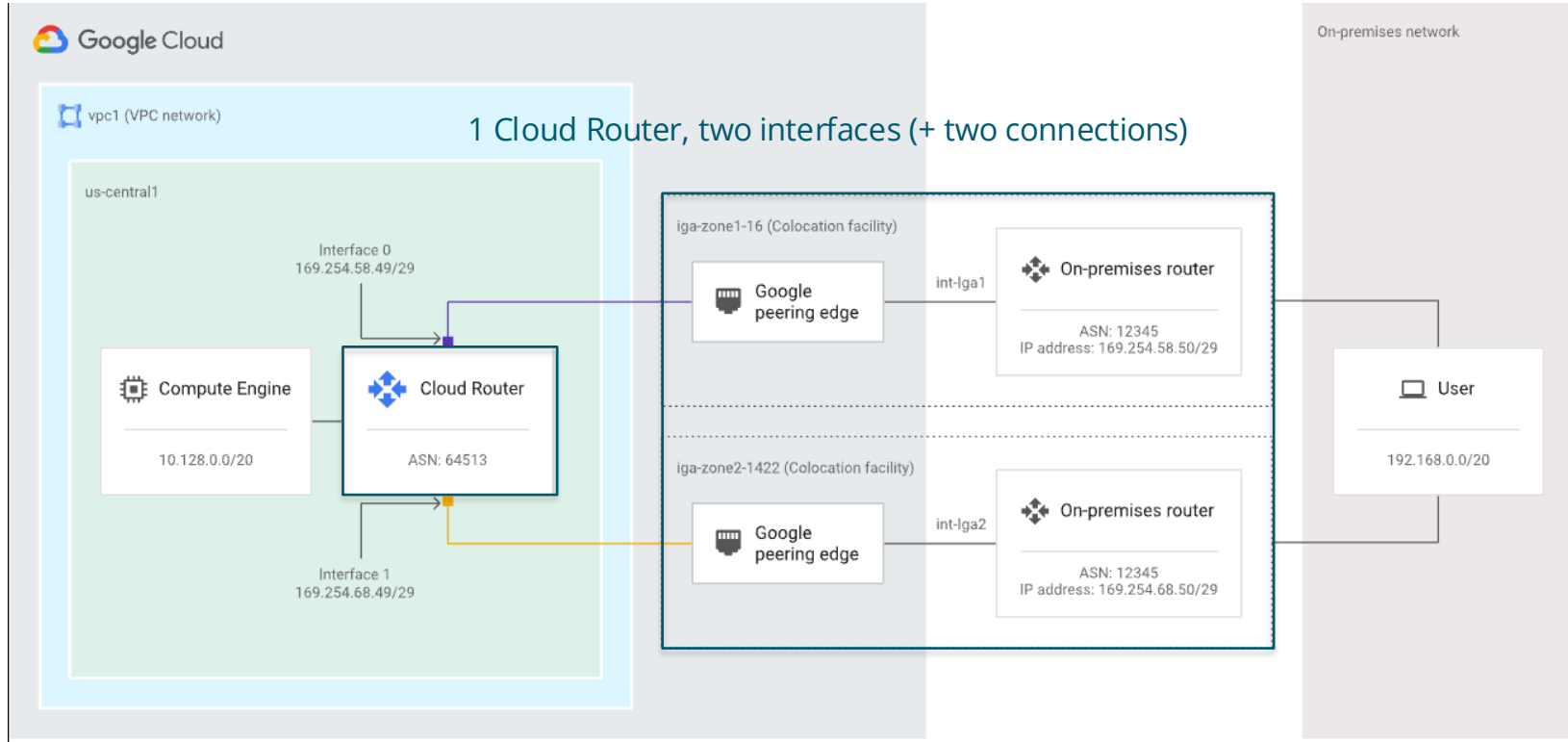
- Disable default networks and configure VPC Firewall
- Secure hybrid and multicloud connectivity
 - IPSec VPN
 - TLS encryption over Interconnect
- Configure Private Service Connect / Private Google Access
- Configure **VPC Service Controls** to secure a perimeter
- Configure **Cloud Armor** to protect against web attacks

Configuring Hybrid and Multi-Cloud Networking: VPN

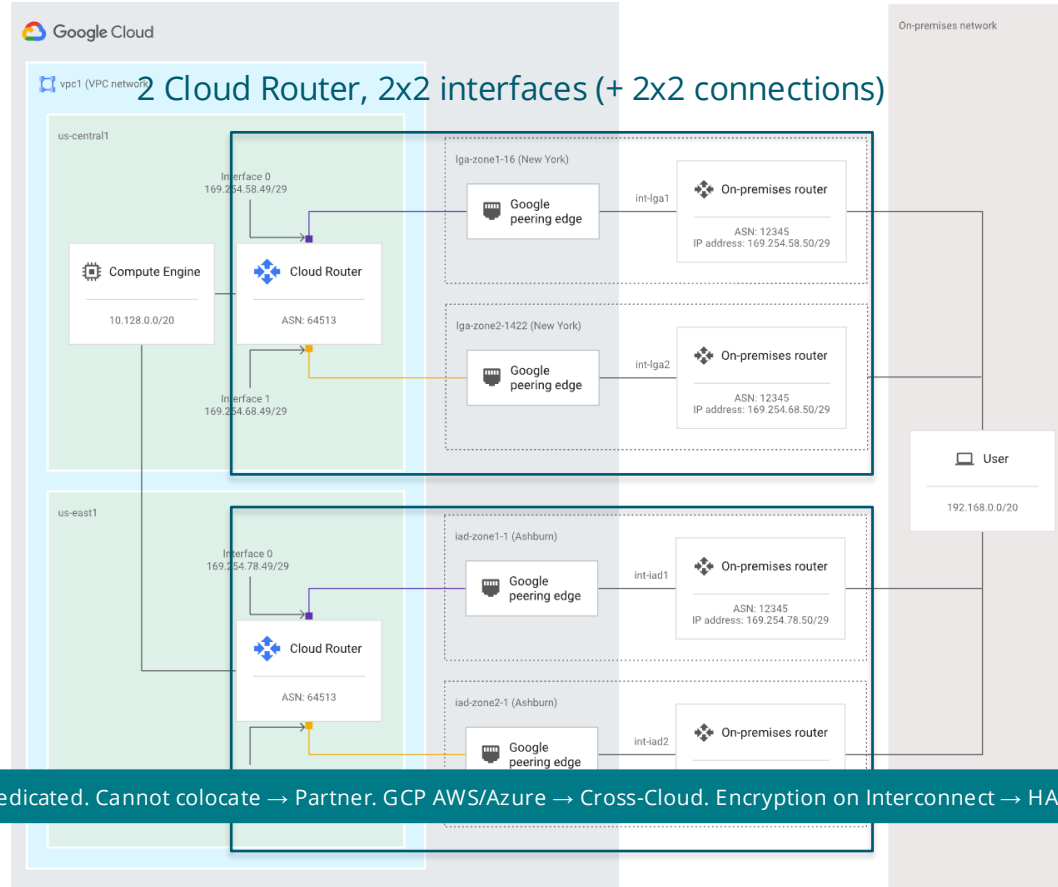


Exam Tip: Need 99.99% SLA → HA VPN. Need IPv6 → HA VPN. Peer device doesn't support BGP → only remaining case for Classic VPN. Encryption over Interconnect → HA VPN over Interconnect.

Configuring Cloud Interconnect (99.9% Topology)



Configuring Cloud Interconnect (99.99% Topology)



Exam Tip: Can colocate + high BW → Dedicated. Cannot colocate → Partner. GCP AWS/Azure → Cross-Cloud. Encryption on Interconnect → HA VPN overlay.

Network Connectivity Center and Multicloud

Network Connectivity Center

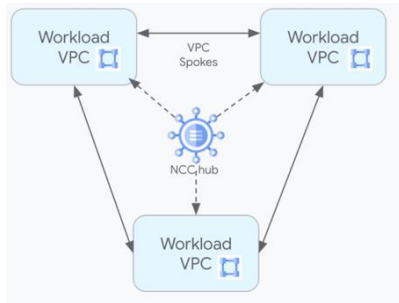
Hub-and-Spoke Orchestration

Spokes attach VPCs, VPN tunnels, Interconnect attachments, or router appliances.

Spoke Types

VPC spokes: replace pairwise peering at scale.

Hybrid spokes: VPN/Interconnect.



Mesh topology

GKE Enterprise (Formerly Anthos)

Fleet Management

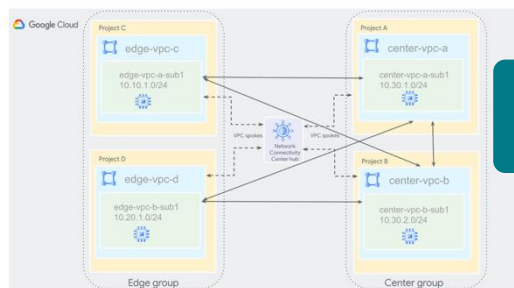
Group and manage K8s clusters across GCP, AWS, Azure, and on-prem from a single control plane.

Key Features

Config Sync: GitOps across fleet.

Policy Controller: OPA-based governance.

Service Mesh: managed Istio.



Star topology

Exam Tip: NCC for transitive VPC routing beyond 25-peering limit.
Star topology to prevent direct workload VPC communication.
GKE Enterprise for multi-cloud K8s management.

Configuring Database Services

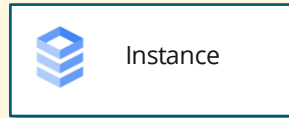
- Standard relational, MySQL/PG migration → **Cloud SQL**
- High-perf PostgreSQL + real-time analytics (4x/100x faster) → **AlloyDB**
- Global SQL transactions, 99.999% → **Cloud Spanner**
- High-throughput NoSQL, >10TB, time-series/IoT → **Bigtable**
- Serverless document DB, mobile/web, offline sync → **Firestore**
- Sub-millisecond caching → **Memorystore** (Valkey preferred for new deployments)

Configuring Relational Database Services

Cloud SQL / AlloyDB

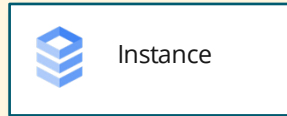
region: us-west1

zone: A

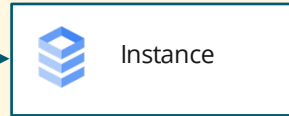


region: us-west1

zone: A



zone: B



Automatic failover

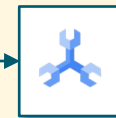
Spanner

region: us-west1

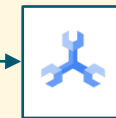
zone: A



zone: B



zone: C



sync

sync

Dual Region: dual-region-india1

Region: asia-south1

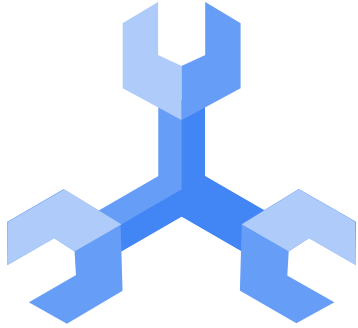


Region: asia-south2



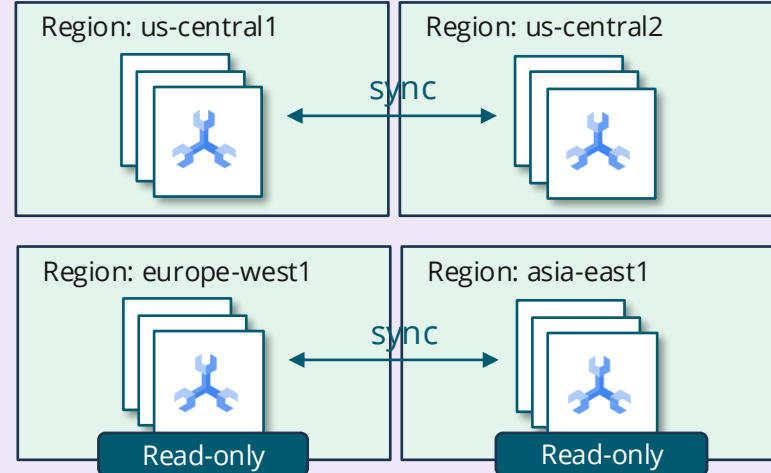
sync

Configuring Relational Database Services



Spanner

Multi Region: nam-eur-asia1



Configuring Relational Database Services

Cloud SQL / AlloyDB

N2

- ☐ 2 vCPU, 16 GB
- ☐ 4 vCPU, 32 GB
- ☒ 8 vCPU, 64 GB
- ☐ 16 vCPU, 128 GB
- ☐ 32 vCPU, 256 GB
- ☐ 48 vCPU, 384 GB
- ☐ 64 vCPU, 512 GB
- ☐ 80 vCPU, 640 GB
- ☐ 96 vCPU, 768 GB
- ☐ 128 vCPU, 864 GB

Spanner



Nodes

Select for large instances.



Processing units (PUs) ?

Select for small, granular sized instances.

Choose a scaling mode



Manual allocation

Set compute capacity for fixed compute resources and costs.

Quantity *

1000

PUs



PUs start at 100, increase by increments of 100 to 1,000 and then by increments of 1,000.

Configuring Relational Database Services

	Cloud SQL	AlloyDB	Spanner
Default backup - frequency	Daily	Daily	Daily
Default backup - number of days	15	14	7
Point-in-time recovery window	Up to 14 (Enterprise) Up to 35 (Enterprise Plus)	Up to 35	Up to 7

Creating a BigQuery Dataset

Source

Create table from

Empty table



Destination

Project *

pluralsight-demo-proj

[Browse](#)

Data set *

demo_dataset

Table *

Maximum name size is 1,024 UTF-8 bytes. Unicode letters, marks, numbers, connectors, dashes and spaces are allowed.

Table type

Native table



Create a BigQuery table for Apache Iceberg

[Preview](#)

Creating a BigQuery Dataset

Source

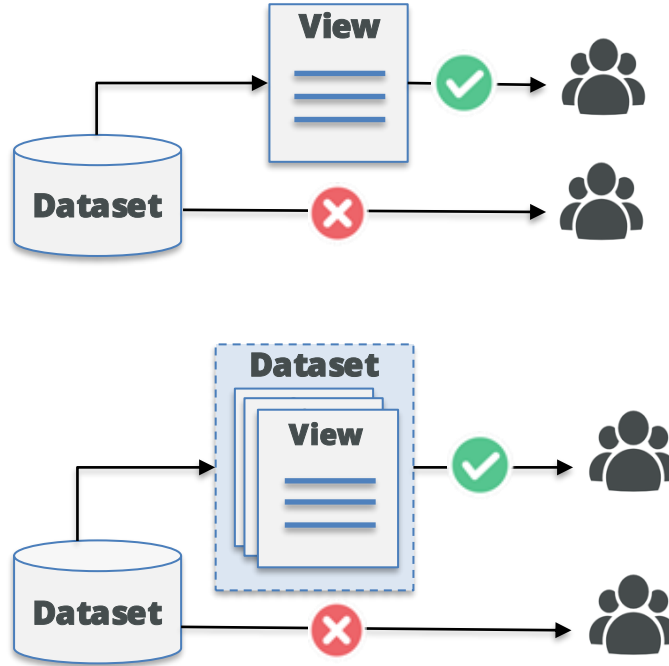
Create table from Empty table

Create table from

- Empty table
- Google Cloud Storage
- Upload
- Drive
- Google Bigtable
- Amazon S3
- Azure Blob storage
- Existing table/view
- Native table

☐ Create a BigQuery table for Apache Iceberg [Preview](#)

Authorizing a BigQuery Dataset



Configuring Firestore

Native mode

A data model oriented around **collections** and **documents**

Real-time updates

Mobile and web client libraries

Scales to millions of concurrent clients

Datastore mode

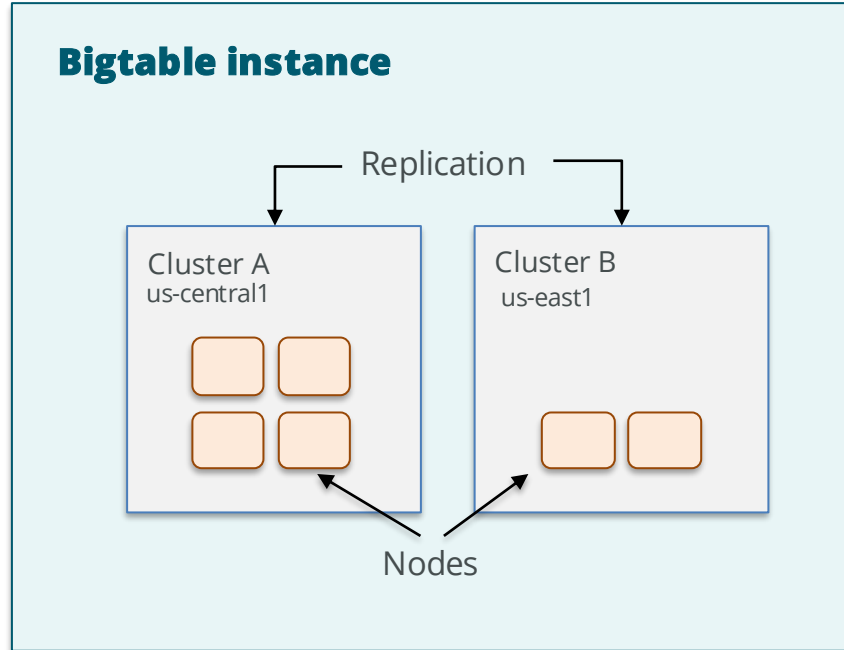
A data model oriented around **entities** and **kinds**

No support for real-time updates

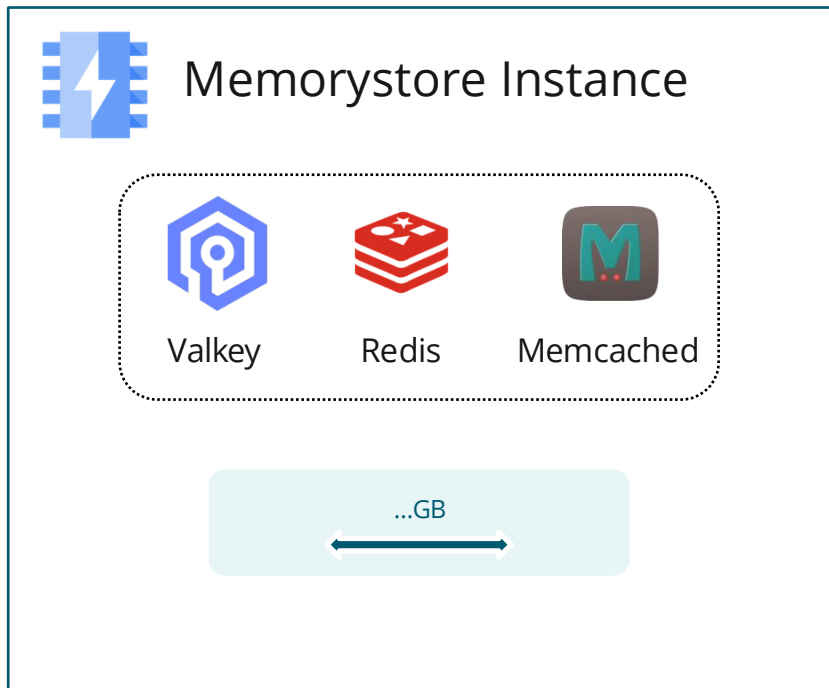
Only server-side client libraries

Scales to millions of writes per second

Configuring Bigtable

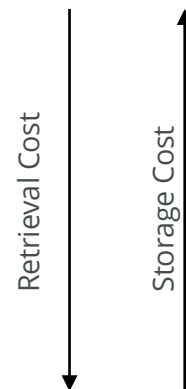


Configuring Memorystore



Storage Classes

Storage class	Minimum duration	Typical monthly availability
Standard Storage	None	>99.99% in multi-regions and dual-regions 99.99% in regions
Nearline Storage	30 days	99.95% in multi-regions and dual-regions 99.9% in regions
Coldline Storage	90 days	99.95% in multi-regions and dual-regions 99.9% in regions
Archive Storage	365 days	99.95% in multi-regions and dual-regions 99.9% in regions



Autoclass

Automatically transitions objects based on access. No retrieval or early deletion fees.

Configuring Cloud Storage

- 1 Choose a **location** (region, dual-region, or multi-region)
- 2 Choose a **storage class** (standard, nearline, coldline, archive) or enable autotclass
- 3 Choose **access control** model (uniform or fine-grained)
- 4 Optionally: add cross-bucket **replication**

Encryption Options

- **Google-managed:** Default, automatic
- **CMEK:** You control keys in Cloud KMS
- **CSEK:** You provide keys per request

Configuring Cloud Storage

Lifecycle Management

- Automate class transitions and deletion
- Conditions: age, createdBefore, storageClass

Retention and Locking

- Bucket Lock: Min retention at bucket level
- Object Lock: Per-object WORM retention

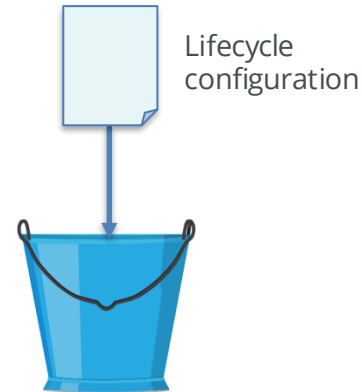
Object Lifecycle Management

- Use cases:
 - Setting a time-to-live (TTL) on objects
 - Retain noncurrent versions of objects
 - Automate “downgrade” of objects’ storage classes
- You set a lifecycle configuration on a bucket

Downgrade the storage class of objects older than 365 days to Coldline storage.

Delete objects created before January 1, 2013.

Keep only the 3 most recent versions of each object in a bucket with versioning enabled.



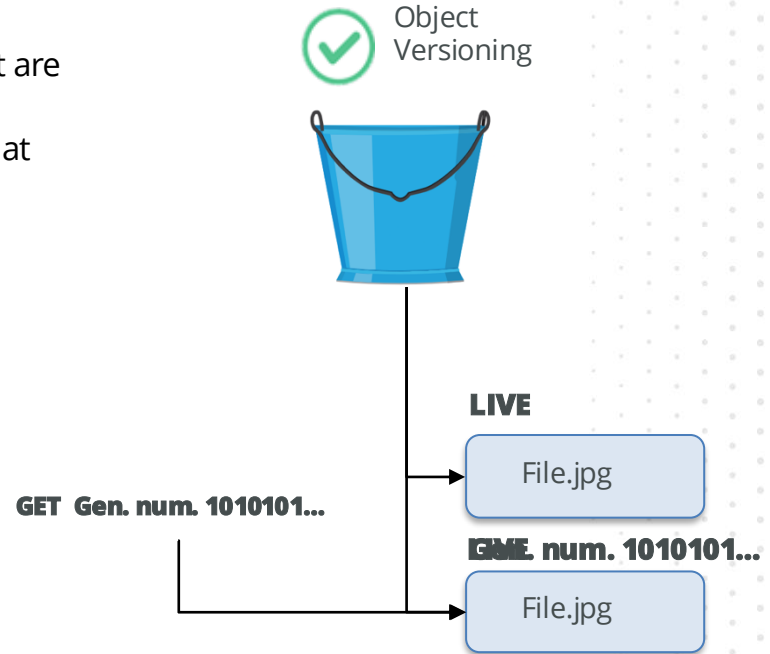
Object Versioning

- Use cases:
 - Support retrieval of objects that are deleted
 - Retrieval of objects that are replaced
- You enable Object Versioning for a bucket



Object Versioning: How it Works

- Cloud Storage retains a noncurrent object version each time you replace or delete a live object version
 - Noncurrent versions retain the same name, but are uniquely identified by generation number
 - Noncurrent versions only appear in requests that explicitly specify the generation number

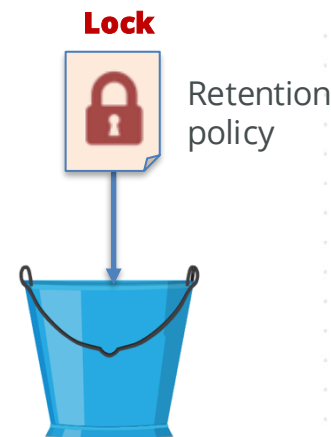


Bucket Lock

- Allows you to configure a data retention policy for a Cloud Storage bucket
- Can help with regulatory and compliance requirements

Bucket Lock: How it Works

- You add a retention policy to a bucket to specify a retention period
 - Objects in the bucket can only be deleted or replaced once their age is greater than the retention period
 - Retention policy retroactively applies to existing objects
- You can lock a retention policy to permanently set it on the bucket
 - You cannot remove the policy or reduce the retention period it has

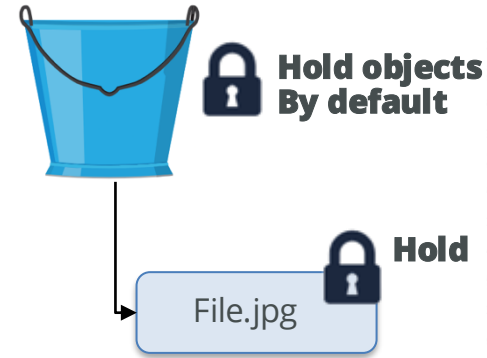


Object Hold

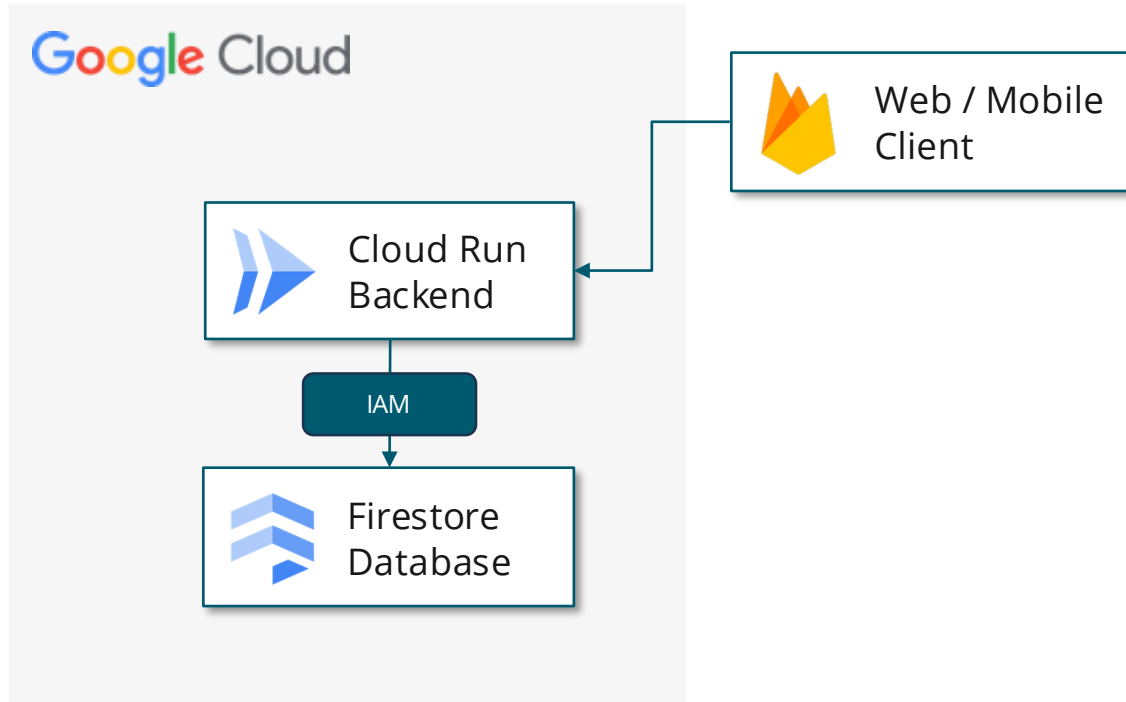
- Holds are object metadata flags
- While an object has a hold, it cannot be deleted or replaced

Object Hold: How it Works

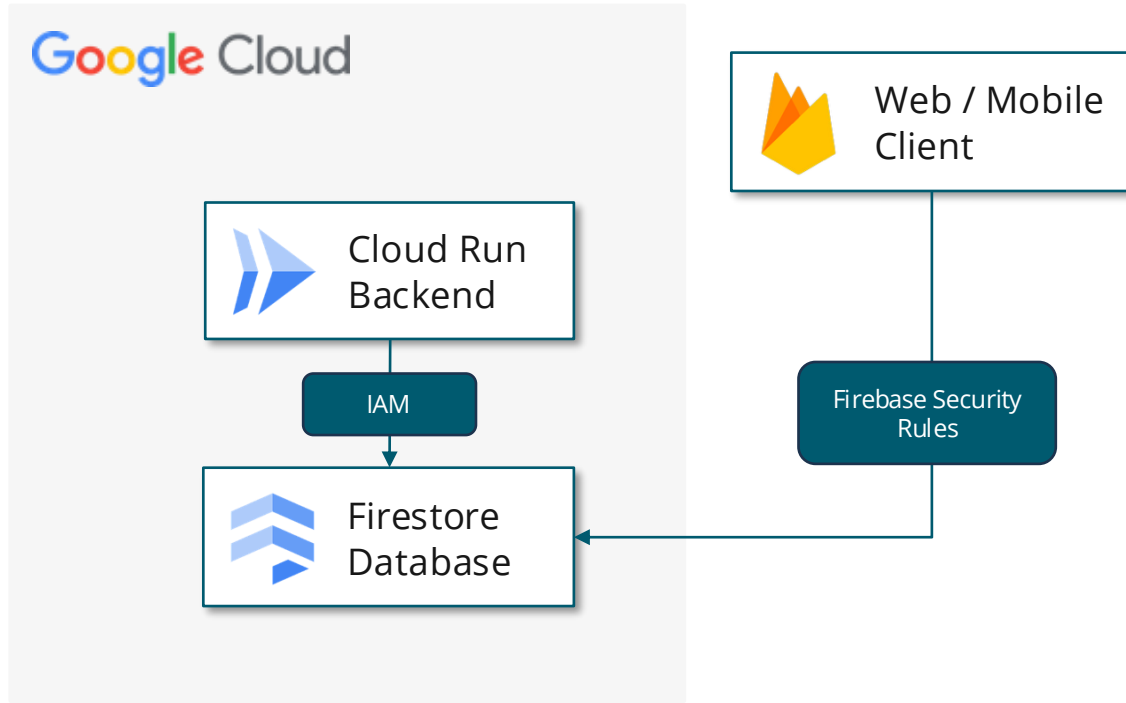
- You place a hold on an object
- You can set the default event-based hold property on a bucket
- A hold can be released any time



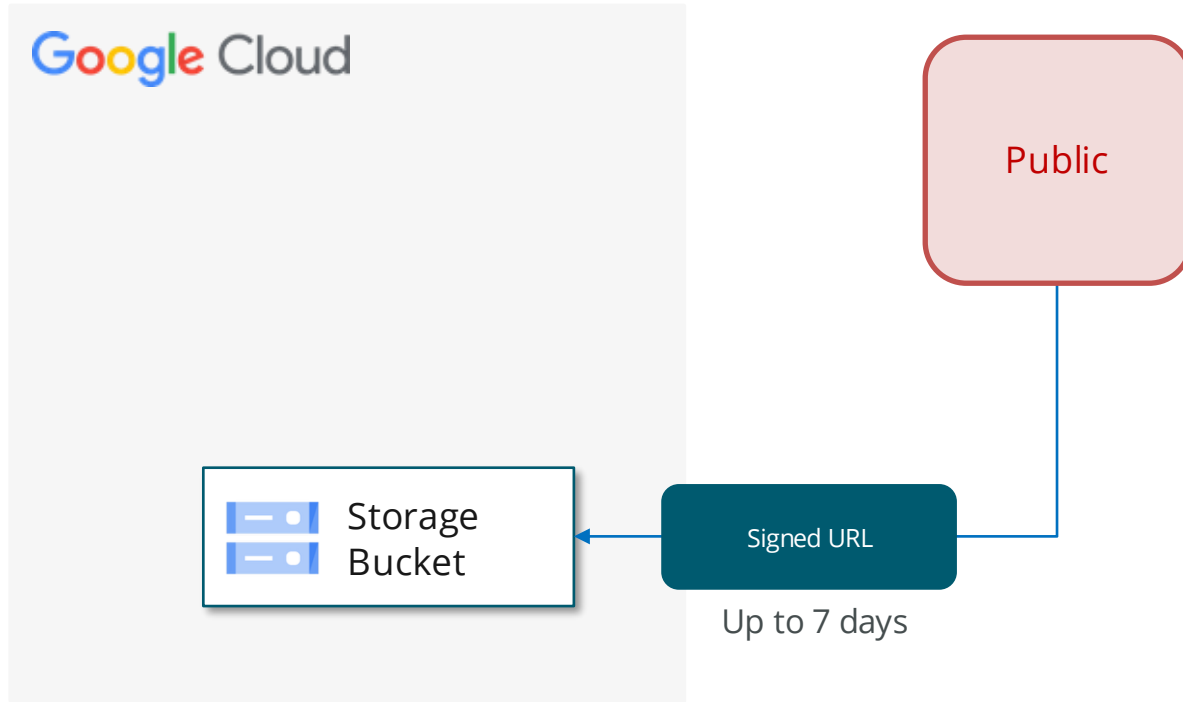
Configuring Firestore Data Access



Configuring Firestore Data Access



Configuring Temporary Access to Objects



Configuring VMs

MIG Autoscaling & VM Manager

Autoscaling Signals

CPU utilization, HTTP LB capacity, Pub/Sub queue depth, Cloud Monitoring custom metrics.

Multiple signals → autoscaler takes the LARGEST recommendation.

Advanced Autoscaling

Predictive: proactive from historical data.

Scheduled: known patterns.

Scale-in controls: min instances, cooldown, max scale-in rate.

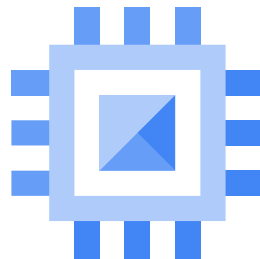
VM Manager

OS patch management at scale. Supports Linux (apt/yum) + Windows. On-demand or scheduled. Patch compliance dashboard. Works with MIGs.

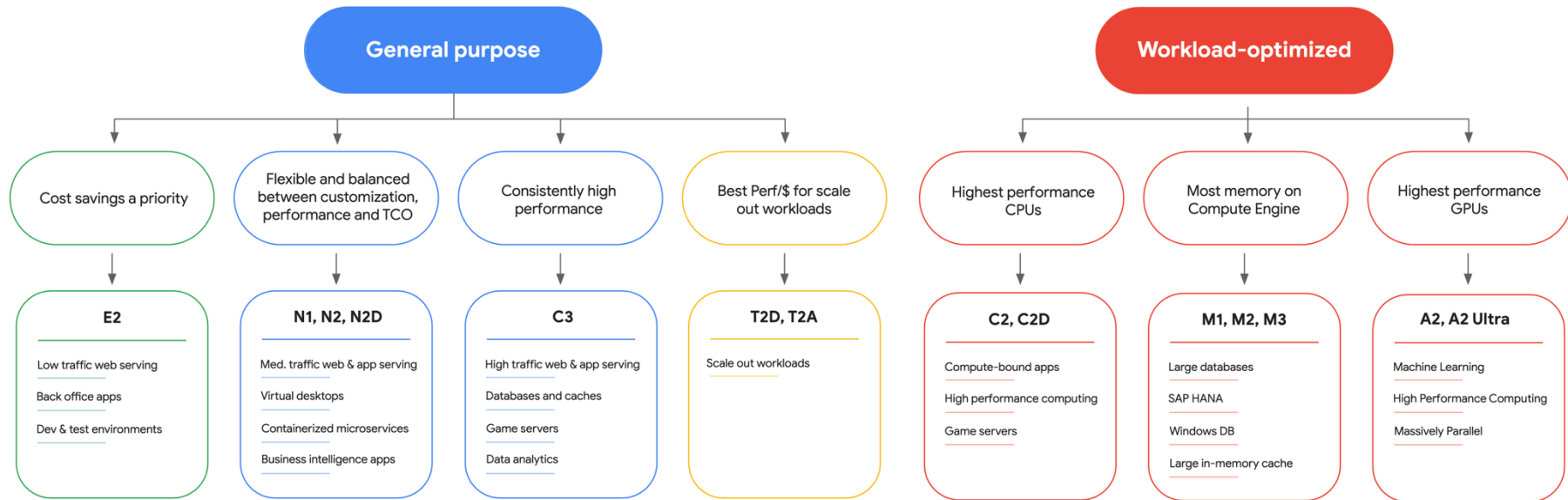
Exam Tip: "Auto-replace failed VMs" → MIG + autohealing.

"Scale on traffic" → MIG + autoscaling. "Patch all VMs" → VM Manager.

Regional MIG for zonal failure protection.



Compute Engine: Machine Families



<https://cloud.google.com/compute>

Creating VMs from Instance Templates

- Quickly create multiple VMs from pre-existing configurations
- Templates pre-define:
 - Machine type
 - Boot disk image
 - Network
 - Labels
 - Management properties

Google Cloud Platform Test-sandbox-td

Create an instance

To create a VM instance, select one of the options:

- New VM instance**
Create a single VM instance from scratch
- New VM instance from template**
Create a single VM instance from an existing template
- Marketplace**
Deploy a ready-to-go solution onto a VM instance

Name ?
instance-2

Region ?
us-east1 (South Carolina)

Machine type
Customize to select cores, memory, and other hardware options
1 vCPU

Container ?
☐ Deploy a container image

Boot disk ?

Provisioning Managed Instance Groups

- Configure a health check to determine if instances are healthy and if they should receive traffic
- **For health checks to work, you must create ingress allow firewall rules so that traffic from Google Cloud probes can connect to your backends.**
- Probes always have IP addresses from a documented range

The screenshot shows the configuration page for a new health check in the Google Cloud console. The form includes fields for Name, Description, Scope (Global/Regional), Protocol (TCP), Port (80), Proxy protocol (NONE), Request, and Response. Below these are the Logs settings (On/Off) and the Health criteria section, which defines the check interval (5 seconds), timeout (5 seconds), healthy threshold (2 consecutive successes), and unhealthy threshold (2 consecutive failures).

Name *

Description

Scope

☒ Global

☐ Regional

Protocol TCP

Port 80

Proxy protocol NONE

Request

Response

Logs

Turning on Health check logs can increase costs in Stackdriver.

☐ On

☒ Off

Health criteria

Define how health is determined: how often to check, how long to wait for a response, and how many successful or failed attempts are decisive

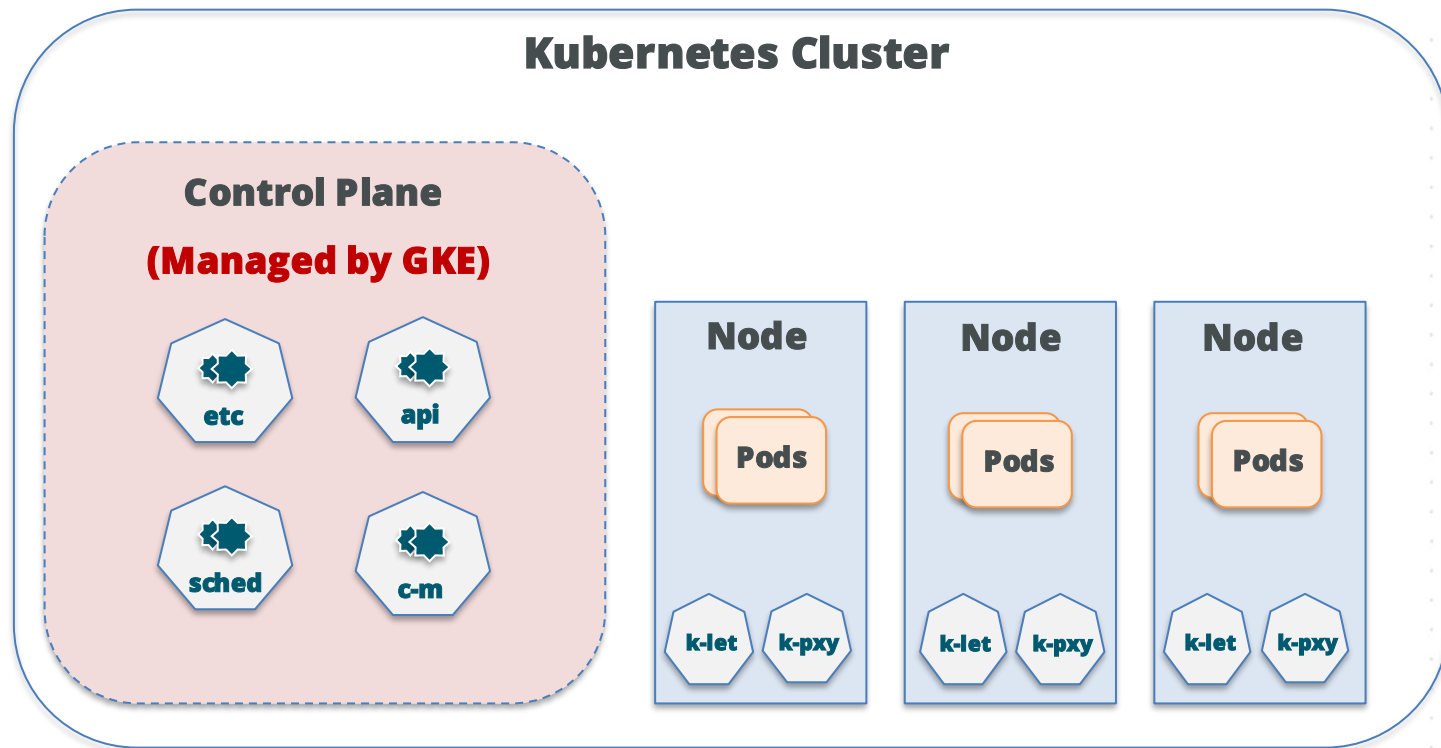
Check interval 5 seconds

Timeout 5 seconds

Healthy threshold 2 consecutive successes

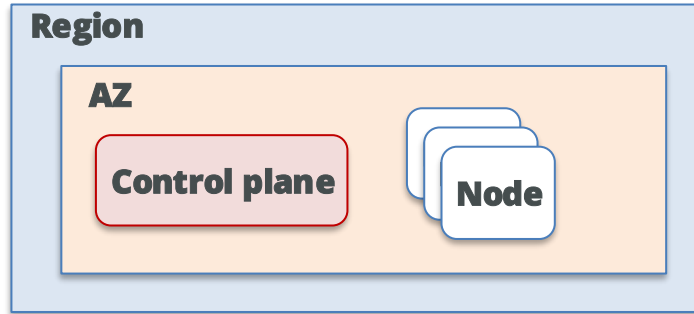
Unhealthy threshold 2 consecutive failures

Kubernetes Cluster Components

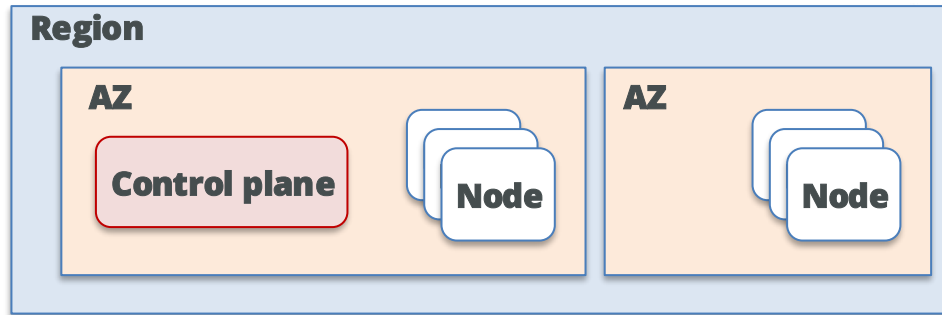


Standard GKE Cluster: Deployment Types

Single-zone cluster

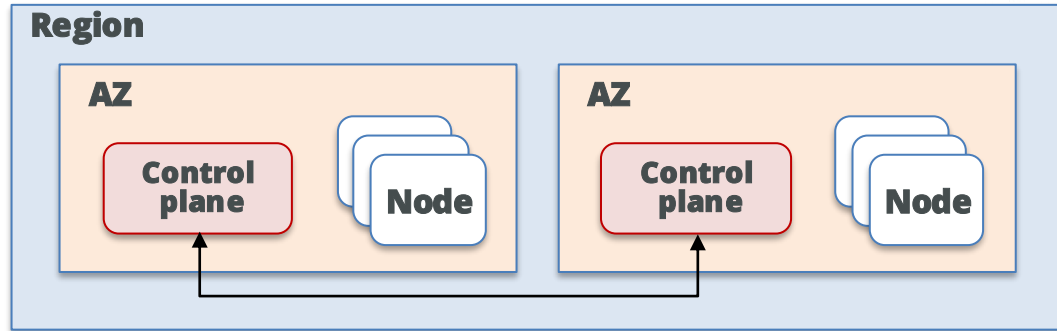


Multi-zonal cluster



Standard GKE Cluster: Deployment Types

Regional cluster



Configuring Cluster Autoscaling

- Enabling autoscaling

```
gcloud container clusters create CLUSTER_NAME \  
  --enable-autoscaling \  
  --num-nodes NUM_NODES \  
  --min-nodes MIN_NODES \  
  --max-nodes MAX_NODES \  
  --region=COMPUTE_REGION
```

Configuring Horizontal Pod Autoscaling

Example YAML

```
apiVersion: autoscaling/v1
kind: HorizontalPodAutoscaler
metadata:
  name: nginx
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: nginx
  minReplicas: 1
  maxReplicas: 10
  targetCPUUtilizationPercentage: 50
```

Configuring Horizontal Pod Autoscaling

Example YAML

```
apiVersion: autoscaling/v1
kind: HorizontalPodAutoscaler
metadata:
  name: nginx
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: nginx
  minReplicas: 1
  maxReplicas: 10
  targetCPUUtilizationPercentage: 50
```

Configuring Horizontal Pod Autoscaling

Example YAML

```
apiVersion: autoscaling/v1
kind: HorizontalPodAutoscaler
metadata:
  name: nginx
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: nginx
  minReplicas: 1
  maxReplicas: 10
  targetCPUUtilizationPercentage: 50
```

Deploying a Stateless App

1) Define a Deployment

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-app
spec:
  replicas: 3
  selector:
    matchLabels:
      run: my-app
  template:
    metadata:
      labels:
        run: my-app
    spec:
      containers:
        - name: hello-app
          image: us-docker.pkg.dev/google-samples/containers/gke/hello-app:1.0
```

Deploying a Stateless App

2) Create a Deployment

```
kubectl apply -f DEPLOYMENT_FILE
```

3) Inspect Deployment

```
kubectl describe deployment DEPLOYMENT_NAME
```

the tool to run
commands against Kubernetes clusters

4) Update/Rollback Deployment

```
kubectl apply -f DEPLOYMENT_FILE
```

```
kubectl rollout undo deployment my-deployment
```

Expose Apps as Services: Service Types

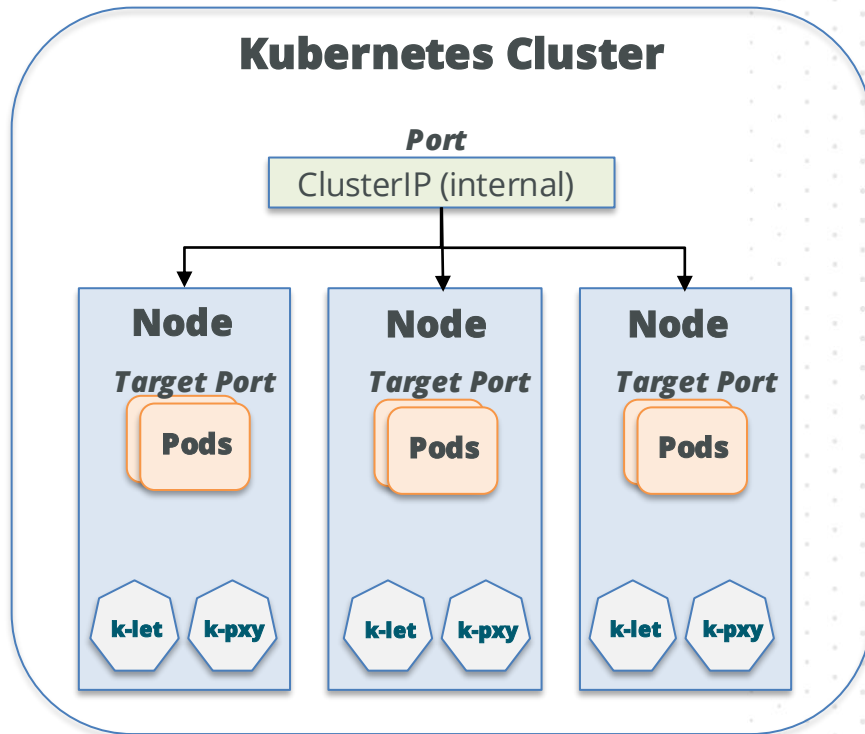
- ClusterIP
- NodePort
- LoadBalancer
- ExternalName

Expose Apps as Services: Service Types

- ClusterIP
- NodePort
- LoadBalancer
- ExternalName

Exposes the Service on a cluster-internal IP

Service is only reachable from within the cluster

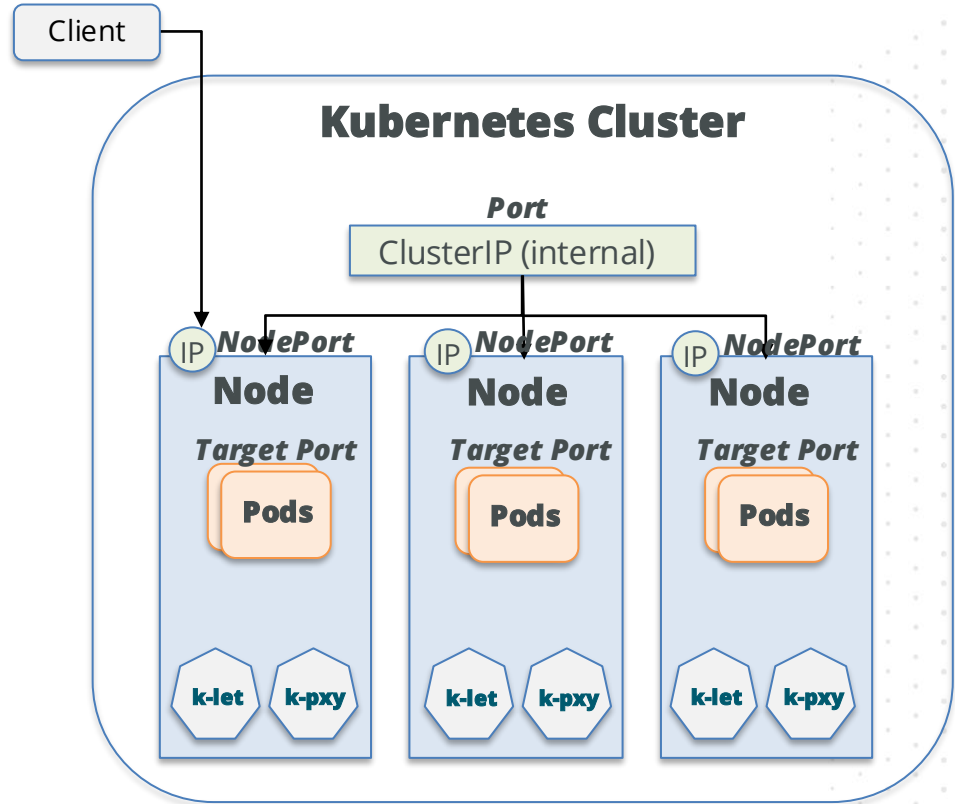


Expose Apps as Services: Service Types

- ClusterIP
- NodePort
- LoadBalancer
- ExternalName

Extension of ClusterIP

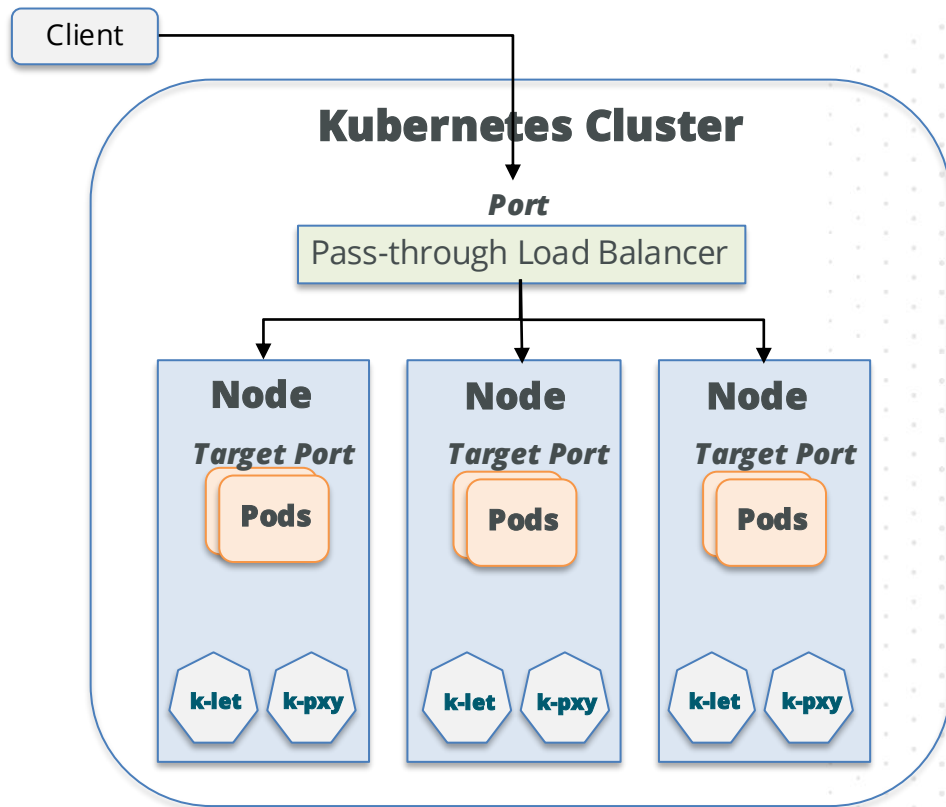
Kubernetes creates a **NodePort** value. The Service is reachable by using the IP address of any node along with the **NodePort** value.



Expose Apps as Services: Service Types

- ClusterIP
- NodePort
- LoadBalancer
- ExternalName

GKE configures an L4, pass-through load balancer (can be internal or external)



Example Service Manifest: LoadBalancer

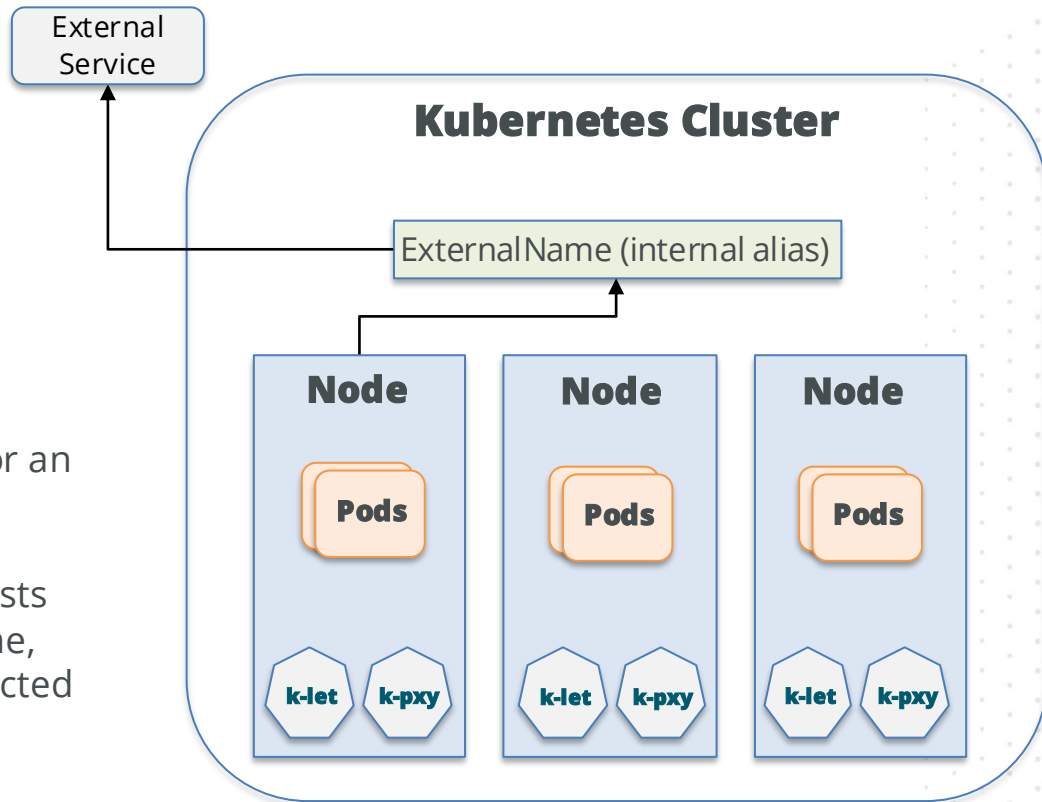
```
apiVersion: v1
kind: Service
metadata:
  name: ilb-service
  annotations:
    networking.gke.io/load-balancer-type: "Internal"
  labels:
    app: hello
spec:
  type: LoadBalancer
  selector:
    app: hello
  ports:
    - port: 80
      targetPort: 8080
      protocol: TCP
```

Expose Apps as Services: Service Types

- ClusterIP
- NodePort
- LoadBalancer
- **ExternalName**

Provides an internal alias for an external DNS name

Internal clients make requests using the internal DNS name, and the requests are redirected to the external name



Load Balancing Traffic with Ingress

- An Ingress object defines rules for routing HTTP(S) traffic to applications running in a cluster
- Backed by the HTTP(S) Load Balancer (internal or external)

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: basic-ingress
spec:
  defaultBackend:
    service:
      name: web
      port:
        number: 8080
---
```

Load Balancing Traffic with Ingress

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: fanout-ingress
spec:
  rules:
  - http:
      paths:
      - path: /*
        pathType: ImplementationSpecific
        backend:
          service:
            name: web
            port:
              number: 8080
      - path: /v2/*
        pathType: ImplementationSpecific
        backend:
          service:
            name: web2
            port:
              number: 8080
```

Choosing How To Expose Apps

ComponentType		Scenarios
Service	ClusterIP	Internal (intra-cluster) access only
Service	NodePort	Need to access the service from outside the cluster For small number of nodes, no load-balancing needed
Service	LoadBalancer	Need to access the service from outside the cluster and to balance the load
Service	ExternalName	Need an internal DNS alias for an external (public) DNS name
Ingress	Internal	Access only on private IP address Need a proxy load balancer and HTTP(S) routing capabilities
Ingress	External	Publicly accessible Need a proxy load balancer and HTTP(S) routing capabilities

GKE Security and Networking

Workload Identity Federation

Maps K8s ServiceAccounts to IAM SAs. Eliminates SA key files. Blocks access to sensitive metadata. Default in Autopilot; enable in Standard. Fleet-wide identity for multi-cluster.

Binary Authorization

Deploy-time control for trusted container images. Attestation policies with image signing. Continuous Validation (CV) for running pods. Integrates with Artifact Registry.

Network Policies

Define pod-to-pod and pod-to-external rules. Default: all pods can communicate. Best practice: global deny-all, then allow specific traffic. Based on pod labels.

Private Clusters

Nodes have only private IPs. Control plane accessible within VPC or via authorized networks. Use bastion host for external API access.

VPC-Native Clusters

Pods get IPs from secondary ranges (alias IPs). Required for container-native load balancing with NEG. Required for internal Ingress (internal ALB).

GKE Sandbox (gVisor)

Additional isolation layer for untrusted code. Intercepts system calls. Reduces attack surface by restricting container capabilities.

Exam Tip: No SA keys in pods → Workload Identity.
Only deploy signed images → Binary Authorization.
Restrict pod communication → Network Policies.
Untrusted code → GKE Sandbox.

Isolate Clusters and Workloads

1) Create a private cluster

```
gcloud container clusters create private-cluster-3 \  
  --create-subnetwork name=my-subnet-3 \  
  --no-enable-master-authorized-networks \  
  --enable-ip-alias \  
  --enable-private-nodes \  
  --master-ipv4-cidr 172.16.0.32/28
```

master-ipv4-cidr specifies an internal IP address range for the control plane

Isolate Clusters and Workloads

1) Create a private cluster

```
gcloud container clusters create private-cluster-3 \  
  --create-subnetwork name=my-subnet-3 \  
  --no-enable-master-authorized-networks \  
  --enable-ip-alias \  
  --enable-private-nodes \  
  --master-ipv4-cidr 172.16.0.32/28
```

Disables authorized networks = any IP address can access the control plane's public endpoint

Isolate Clusters and Workloads

2) Restrict control plane access with *authorized networks*

Specify authorized networks:

```
gcloud container clusters update private-cluster-1 \  
  --enable-master-authorized-networks \  
  --master-authorized-networks 203.0.113.0/29
```

Sources from 203.0.113.0/29 have access to the control plane

Isolate Clusters and Workloads

Alternatively, disable control plane public endpoint

```
gcloud container clusters create private-cluster-3 \  
  --create-subnetwork name=my-subnet-3 \  
  --no-enable-master-authorized-networks \  
  --enable-ip-alias \  
  --enable-private-nodes \  
  --enable-private-endpoint \  
  --master-ipv4-cidr 172.16.0.32/28
```

Control plane will only be accessible from within the VPC

Vertex AI Pipelines and MLOps

Vertex AI provides a unified platform for the full ML lifecycle: data prep, training, evaluation, deployment, and monitoring.

Vertex AI Pipelines

ML workflow as DAG of containerized tasks. Supports Kubeflow Pipelines SDK and TFX SDK. Managed execution — no infra provisioning. Template Gallery for bootstrapping. CI/CD via Cloud Build or GitHub Actions.

Model Registry

Central hub for model versions and lineage. Model cards for documentation. Deploy, evaluate, or batch inference directly. Discoverable across regions/projects via Dataplex catalog.

Experiments & Metadata

Compare model performance side-by-side. Track code, data, hyperparameters, results. ML Metadata records lineage for reproducibility and audit.

Feature Store

Managed feature store built on BigQuery. Online serving via Bigtable or Optimized serving. Supports embeddings + vector similarity search. Share and reuse features across teams. Use gRPC API for best performance.

MLOps Lifecycle

1. Data (BigQuery/Dataflow)
2. Features (Feature Store)
3. Train (AutoML/Custom)
4. Evaluate
5. Register (Model Registry)
6. Deploy
7. Monitor (drift detection)

Exam Tip: Automate ML workflow → Pipelines.
Share features across teams → Feature Store.
Track model lineage → ML Metadata.
Version models → Model Registry.

Vertex AI Prediction and Model Monitoring

Online vs Batch Prediction

- Batch prediction at **50% discount** vs online
- Gemini batch inference also at 50% discount
- Dedicated endpoints: streaming, gRPC, OpenAI-compatible

Model Monitoring

What It Monitors

Prediction drift: output patterns changing.

Feature drift: input data distribution shifting.

Model degradation: declining performance metrics.

MLOps Loop

When drift detected → trigger pipeline rerun for retraining.
Alerting + diagnosis + feature attribution explanations

Exam Tip: Real-time predictions → Online endpoint.
Score millions overnight → Batch (50% savings).
Detect performance degradation → Model Monitoring.
Drift → retrain via Pipeline.

Questions Breakdown

A financial services company needs to connect its on-premises data center to Google Cloud. They require 99.99% availability SLA for this connectivity, with 10 Gbps bandwidth, and IPSec encryption. The company is colocated at a Google PoP.

Which connectivity solution should you recommend?

- A.** HA VPN with two tunnels using ECMP for load balancing
- B.** Dedicated Interconnect with HA VPN overlay for encryption
- C.** Partner Interconnect with two VLAN attachments in different edge availability domains
- D.** Dedicated Interconnect with MACsec encryption enabled

Questions Breakdown

A financial services company needs to connect its on-premises data center to Google Cloud. They require 99.99% availability SLA for this connectivity, with 10 Gbps bandwidth, and IPSec encryption. The company is colocated at a Google PoP.

- A. HA VPN with two tunnels using ECMP for load balancing
- B. Dedicated Interconnect with HA VPN overlay for encryption**
- C. Partner Interconnect with two VLAN attachments in different edge availability domains
- D. Dedicated Interconnect with MACsec encryption enabled

Questions Breakdown

You set up an HTTP(S) load balancer and health check with a managed instance group (MIG) configured as a backend service. You noticed that, after deploying the load balancer, MIG instances are being terminated and re-launched every 60 seconds.

What should you do?

- A. Configure an ingress VPC firewall rule to allow source traffic to reach the HTTP(S) load balancer
- B. Configure a public IP address on each instance and an ingress VPC firewall rule to allow source traffic to reach the instances
- C. Configure an ingress VPC firewall rule to allow load balancer health checks to reach the instances
- D. Ensure the default "allow all" egress rule is enforced in the VPC firewall

Questions Breakdown

You set up an HTTP(S) load balancer and health check with a managed instance group (MIG) configured as a backend service. You noticed that, after deploying the load balancer, MIG instances are being terminated and re-launched every 60 seconds.

What should you do?

- A. Configure an ingress VPC firewall rule to allow source traffic to reach the HTTP(S) load balancer
- B. Configure a public IP address on each instance and an ingress VPC firewall rule to allow source traffic to reach the instances
- C. Configure an ingress VPC firewall rule to allow load balancer health checks to reach the instances
- D. Ensure the default "allow all" egress rule is enforced in the VPC firewall

Questions Breakdown

A healthcare company stores sensitive patient data in BigQuery and Cloud Storage. A recent audit found that an employee leaked sensitive data by copying it to an unauthorized project. You need to prevent data exfiltration while allowing access only from corporate VPN IP ranges. The solution must work across multiple projects.

Which solution should you implement?

- A.** Configure IAM Deny policies to block cross-project data access
- B.** Create a VPC Service Controls perimeter with access levels restricted to corporate IP ranges
- C.** Enable VPC Flow Logs and create Cloud Monitoring alerts for unusual data transfers
- D.** Use Organization Policy constraints to restrict resource sharing to specific projects

Questions Breakdown

A healthcare company stores sensitive patient data in BigQuery and Cloud Storage. A recent audit found that an employee leaked sensitive data by copying it to an unauthorized project. You need to prevent data exfiltration while allowing access only from corporate VPN IP ranges. The solution must work across multiple projects.

Which solution should you implement?

- A. Configure IAM Deny policies to block cross-project data access
- B. Create a VPC Service Controls perimeter with access levels restricted to corporate IP ranges**
- C. Enable VPC Flow Logs and create Cloud Monitoring alerts for unusual data transfers
- D. Use Organization Policy constraints to restrict resource sharing to specific projects

Questions Breakdown

You are responsible for deploying a container-based secure web application to Google Kubernetes Engine (GKE). You need to ensure the application scales automatically and that traffic can be served to end users over HTTPS. How should you configure the GKE deployment.

How should you configure the GKE deployment?

- A.** Configure an autoscaling policy for the underlying Compute Engine instance group. Define an Ingress resource
- B.** Configure an autoscaling policy for the underlying Compute Engine instance group. Define a Service resource of type LoadBalancer
- C.** Enable cluster autoscaling and use the Horizontal Pod Autoscaler. Define a Service resource of type LoadBalancer
- D.** Enable cluster autoscaling and use the Horizontal Pod Autoscaler. Define an Ingress resource

Questions Breakdown

You are responsible for deploying a container-based secure web application to Google Kubernetes Engine (GKE). You need to ensure the application scales automatically and that traffic can be served to end users over HTTPS. How should you configure the GKE deployment.

How should you configure the GKE deployment?

- A. Configure an autoscaling policy for the underlying Compute Engine instance group. Define an Ingress resource
- B. Configure an autoscaling policy for the underlying Compute Engine instance group. Define a Service resource of type LoadBalancer
- C. Enable cluster autoscaling and use the Horizontal Pod Autoscaler. Define a Service resource of type LoadBalancer
- D. Enable cluster autoscaling and use the Horizontal Pod Autoscaler. Define an Ingress resource**

Questions Breakdown

You are configuring a managed relational database solution on Google Cloud for an application that requires the ability to revert its state to any specific point in time within the last 7 days to recover from potential operational errors. Daily backups must be automated, and the backup data should be stored durably.

What should you do?

- A.** Configure Cloud SQL with point-in-time recovery enabled and a 7-day transaction log retention. Ensure automated daily backups are configured, utilizing a multi-region storage for backups
- B.** Deploy a MySQL database to a Compute Engine VM. Set up a snapshot schedule for the VM's persistent disk to run daily, retaining snapshots for 7 days
- C.** Configure Cloud SQL with the option "create failover replicas" for high availability and configure automated backups with a 7-day retention period
- D.** Deploy a MySQL database to a Compute Engine VM. Automate daily database exports to a Cloud Storage bucket, retaining these exports for 7 days

Questions Breakdown

You are configuring a managed relational database solution on Google Cloud for an application that requires the ability to revert its state to any specific point in time within the last 7 days to recover from potential operational errors. Daily backups must be automated, and the backup data should be stored durably.

What should you do?

- A.** Configure Cloud SQL with point-in-time recovery enabled and a 7-day transaction log retention. Ensure automated daily backups are configured, utilizing a multi-region storage for backups
- B.** Deploy a MySQL database to a Compute Engine VM. Set up a snapshot schedule for the VM's persistent disk to run daily, retaining snapshots for 7 days
- C.** Configure Cloud SQL with the option "create failover replicas" for high availability and configure automated backups with a 7-day retention period
- D.** Deploy a MySQL database to a Compute Engine VM. Automate daily database exports to a Cloud Storage bucket, retaining these exports for 7 days

Questions Breakdown

Your team deployed a fraud detection model on Vertex AI. After two months in production, the model's accuracy has decreased significantly. Investigation reveals that transaction patterns have shifted due to a new payment method. You need to detect such degradation automatically in the future and trigger model retraining without manual intervention.

What should you do?

- A.** Schedule weekly batch predictions and compare results against a validation dataset
- B.** Enable Vertex AI Model Monitoring with feature drift detection and trigger training pipeline on alert
- C.** Deploy a shadow model and use A/B testing to compare performance continuously
- D.** Use Cloud Monitoring custom metrics to track prediction latency as a proxy for model health

Questions Breakdown

Your team deployed a fraud detection model on Vertex AI. After two months in production, the model's accuracy has decreased significantly. Investigation reveals that transaction patterns have shifted due to a new payment method. You need to detect such degradation automatically in the future and trigger model retraining without manual intervention.

What should you do?

- A. Schedule weekly batch predictions and compare results against a validation dataset
- B. Enable Vertex AI Model Monitoring with feature drift detection and trigger training pipeline on alert**
- C. Deploy a shadow model and use A/B testing to compare performance continuously
- D. Use Cloud Monitoring custom metrics to track prediction latency as a proxy for model health

Segment 6: Designing for Security, and Compliance



Objectives

- Identity and Access Management (IAM)
- Resource hierarchy
- Service controls
- Securing data at rest and in transit
- Secrets and certificate management
- Compliance considerations
- Securing AI

The Principle of Least Privilege

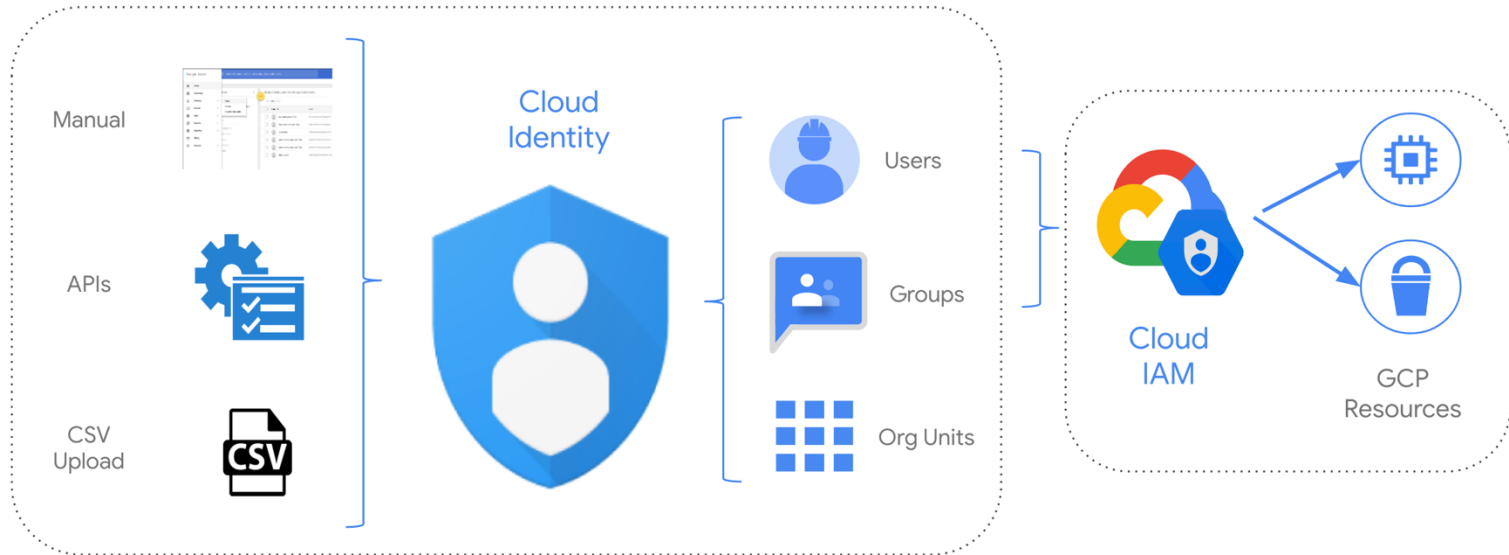
A resource/individual should only have access to the exact resource(s) it needs to function

Authentication and Authorization

Add/Import

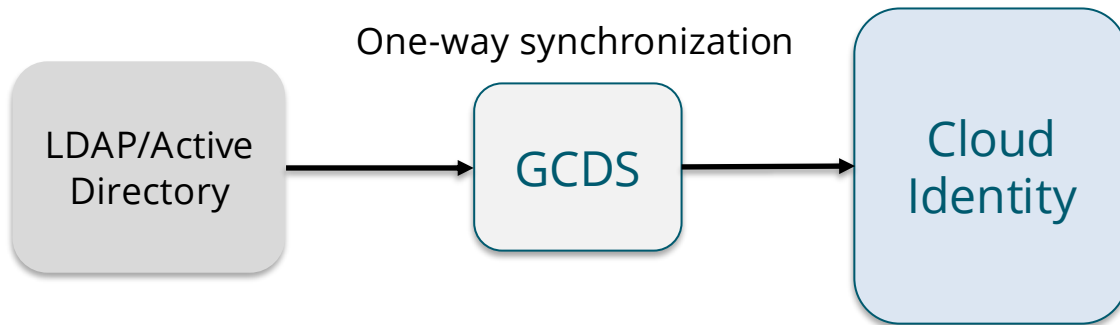
Manage users and group membership

Manage access controls on GCP



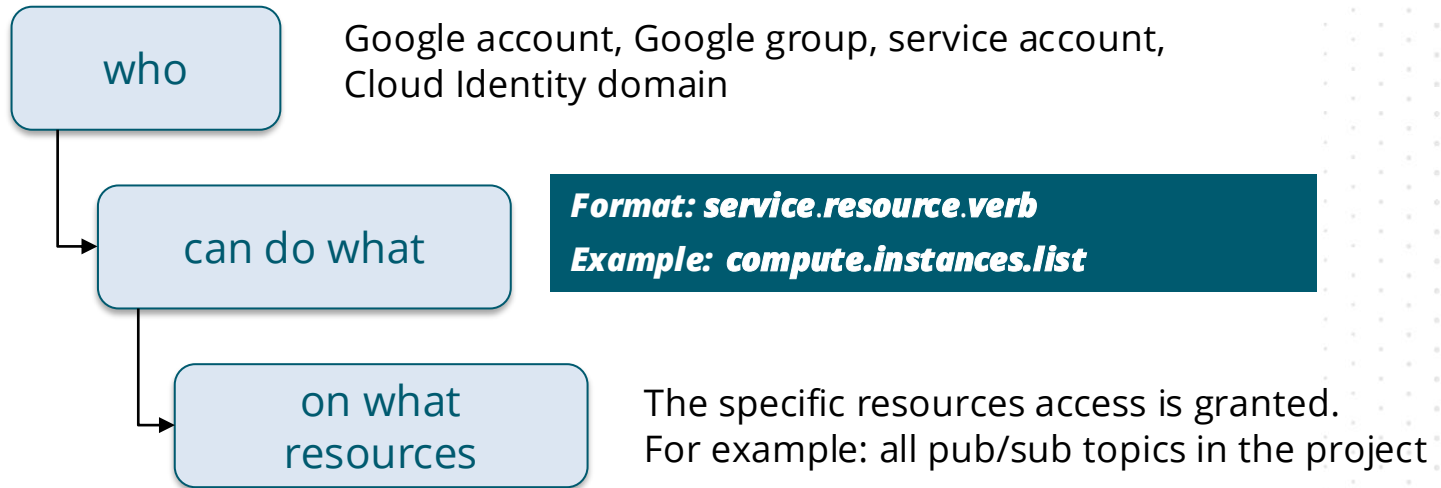
Cloud Identity: Directory Sync

Cloud Identity supports extending an on-premises directory to the cloud with **Google Cloud Directory Sync (GCDS)**



Authorization on GCP: Cloud IAM

- Consistent access control interface for *all* Google Cloud services
- An IAM policy can be broken into three components:



Cloud IAM Roles

Roles

- **Basic roles:** Owner, Editor, Viewer
- **Predefined roles:** service-specific roles (example: Pub/Sub Subscriber)
- **Custom roles:** based on user-specified list of permissions (can only be set at the *project* or *organization* level)

IAM Recommender

Analyzes 90 days of permission usage.

Suggests switching to tighter predefined roles or removing unused bindings.

Provides a confidence score and impact analysis before applying.

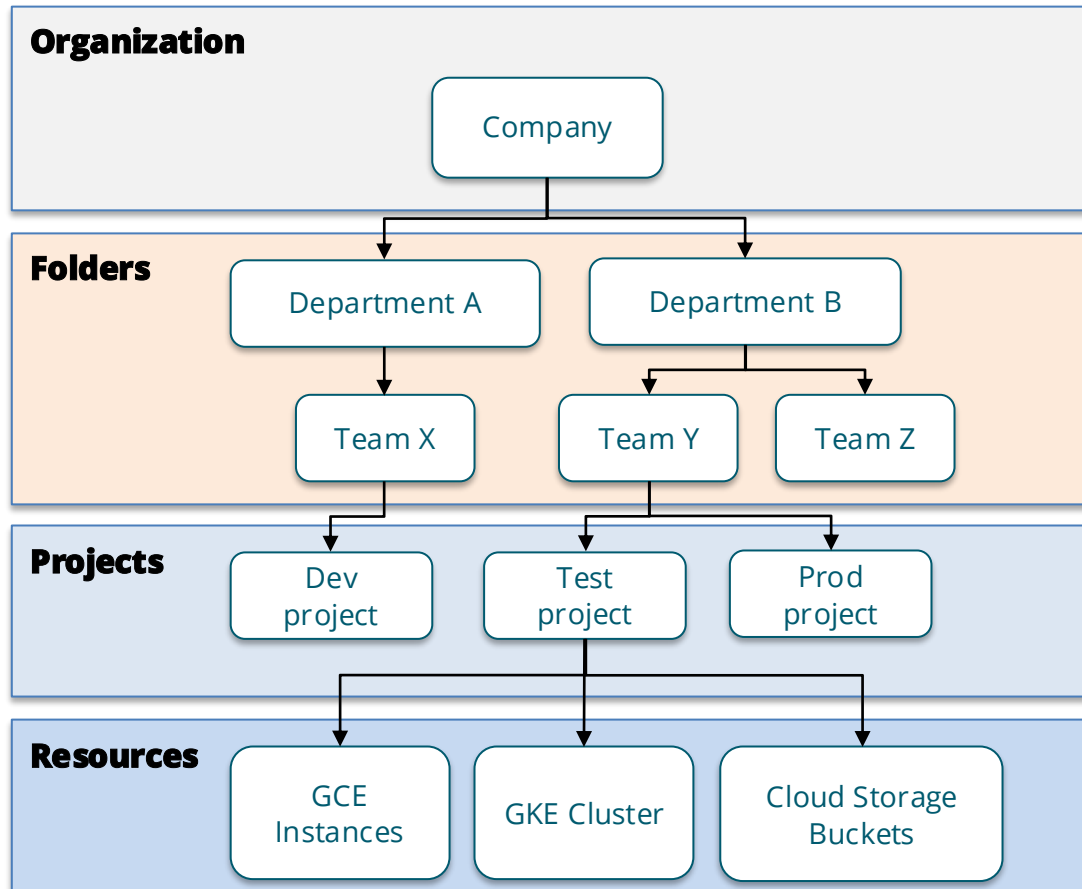
Service Accounts

- Associated with a public/private RSA key pair (no password)
- Both a *resource* and an *identity*
 - You can grant a role to a service account (identity)
 - You can grant roles to users to access (impersonate) or manage the service account (resource)

Enforcing Least Privilege: Best Practices

- Do not grant basic roles ("Owner", "Editor", "Viewer")
- Use role recommendations and grant roles at the smallest scope needed
- Grant roles to Google groups instead of individuals
- Be cautious when granting the Service Account User role
- Regularly check Cloud Audit Logs

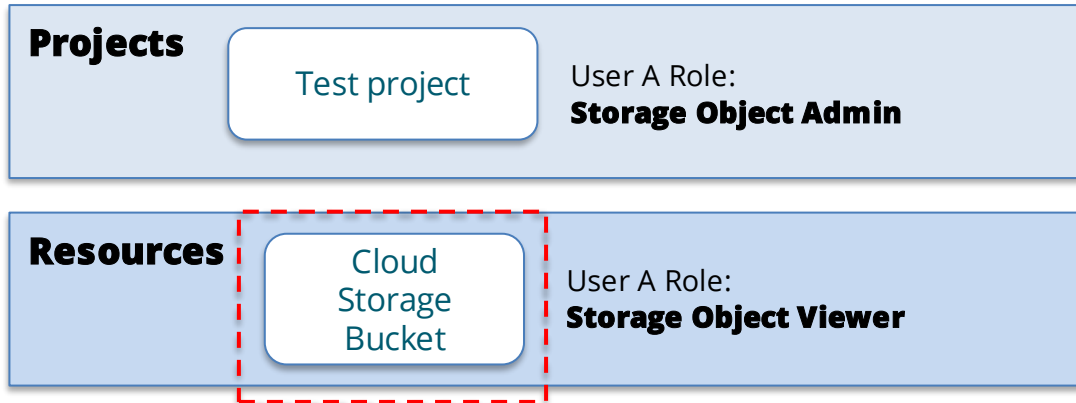
Resource Hierarchy



Using Resource Hierarchy for Access Control

- IAM roles can be set at the organization level, folder level, project level, or (in some cases) the resource level
- IAM roles granted at a level are inherited by all resources under that level
- The effective *allow* policy for a resource is the **union** of the inherited *allow* policy and the resource *allow* policy
- **There is no way to explicitly remove a permission for a lower-level resource that is granted at a higher level in the resource hierarchy**

Example Policy Inheritance



User A final policy = Parent policy **U** Resource policy
Storage Object Admin **U** Storage Object Viewer
Final Access scope = **Storage Object Admin**

Organization Policy Service

- Centralized and programmatic enforcement of constraints across the entire resource hierarchy
- Define and establish guardrails for teams to stay within compliance boundaries

Example Policy Constraints

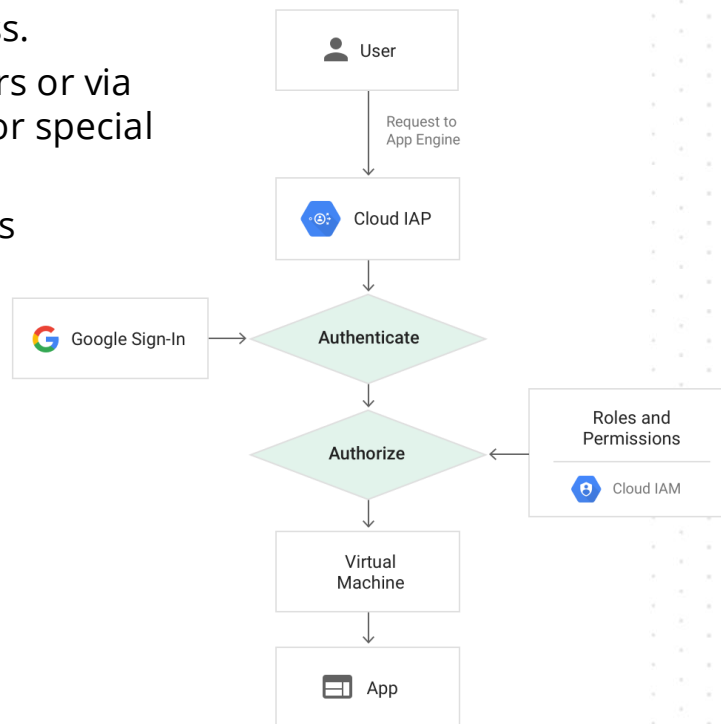
Constraint	Description	Example
Google Cloud Platform – Resource Location Restriction	Defines the set of locations where location-based GCP resources can be created	<i>in:us-locations</i>
Restrict Resource Service Usage	Defines the set of Google Cloud resource services that can be used within an organization, folder, or project	<i>is:compute.googleapis.com,</i> <i>is:storage.googleapis.com</i>

Example Policy Constraints

Constraint	Description	Example
Compute Engine – Define allowed external IPs for VM instances	Defines the set of Compute Engine VM instances that are allowed to use external IP addresses.	<i>projects/myproject1234/zones/us-east1-c/instances/INSTANC E1</i>
Cloud SQL – Restrict Public IP access on Cloud SQL instances	Restricts the configuration of public IPs on Cloud SQL instances	<i>True</i>

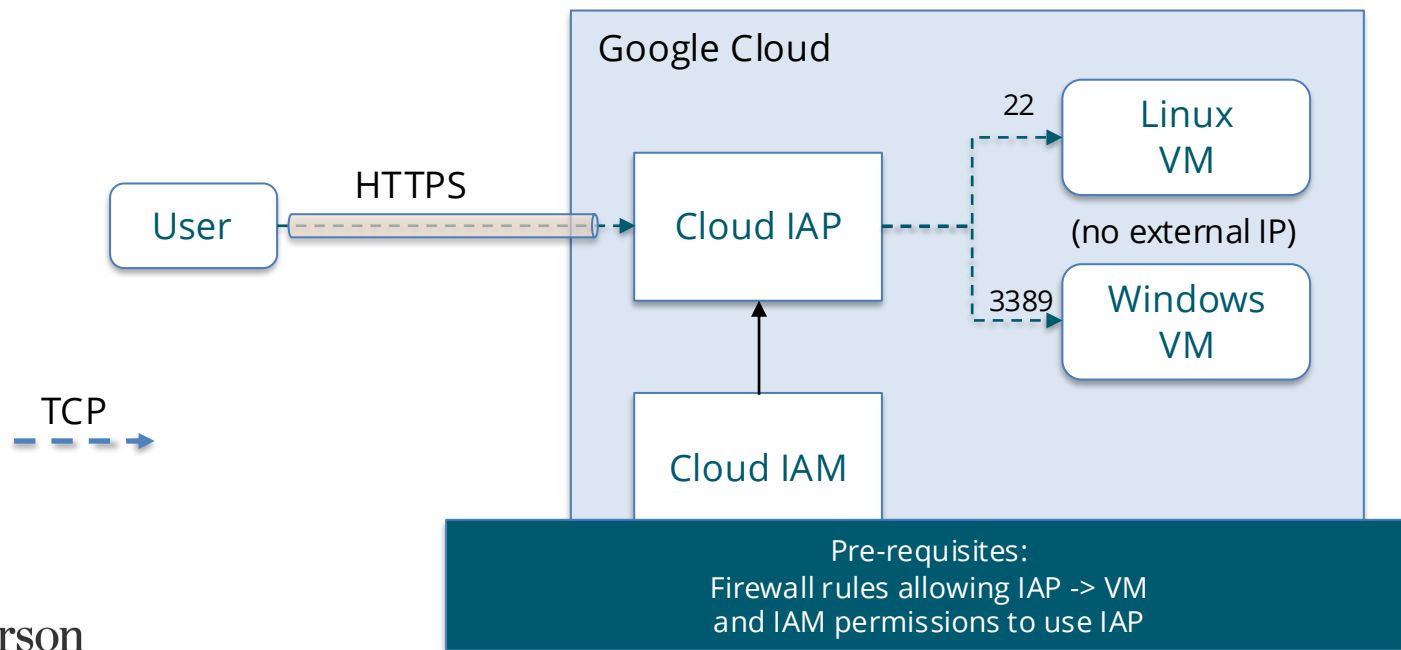
Identity-Aware Proxy (IAP)

- Sits in front of web apps and VMs. Verifies user identity and context before granting access.
- Protects apps behind HTTPS load balancers or via IAP TCP forwarding for SSH/RDP. No VPN or special client needed for web apps
- Central authorization layer for applications



Connecting to VMs using IAP TCP Forwarding

For VMs without external IP address, you can enable **Identity-Aware Proxy TCP forwarding**



Security Command Center

SCC is Google Cloud's security and risk management platform.

It provides **asset inventory**, **vulnerability detection**, **threat detection**, and **compliance monitoring** in a single pane of glass.

Security Health Analytics

Scans for misconfigurations: public buckets, open firewall rules, unencrypted disks, overly permissive IAM. Maps findings to compliance frameworks (CIS, PCI-DSS, NIST, ISO 27001).



Event Threat Detection

Monitors Cloud Logging in near-real time. Detects: brute force SSH, cryptocurrency mining, IAM anomalies, exfiltration attempts, malware. Uses ML + threat intelligence. Consumes VPC Flow Logs automatically.

Container Threat Detection

Monitors GKE nodes via DaemonSet. Detects: reverse shells, malicious scripts, binary execution anomalies. File monitoring detectors are disabled by default (performance impact on CI/CD workloads).

Web Security Scanner

Scans App Engine, GKE, and Compute Engine web apps. Detects XSS, SQL injection, mixed content, outdated libraries. Supports both managed and custom scans.

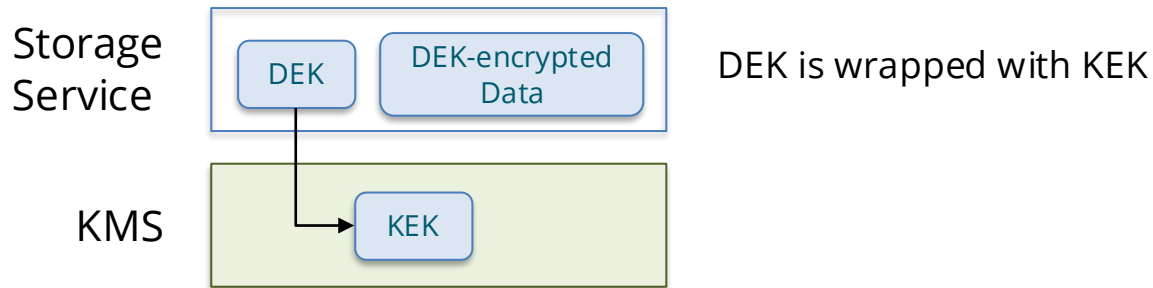
Exam Tip: "Detect crypto mining" → Event Threat Detection
"Find misconfigured buckets" → Security Health Analytics.
"CIS compliance dashboard" → SCC Premium.

Cloud Audit Logs

- Contains the logs that answer the question *"Who did what, where, and when?"*
 - Admin Activity logs (**can't be disabled**)
 - System Event logs (**can't be disabled**)
 - Data Access logs (must be enabled)

Data Encryption on GCP

Google Cloud always encrypts data on the server side (**AES256**), before it is written to disk



Google Default Encryption

Multi-layer key hierarchy: DEK → KEK → KMS master key → Root KMS.
Automatic key rotation. Zero customer configuration needed. AES-256 for all data at rest.

Data Encryption on GCP: CMEK, CSEK

CMEK (Customer-Managed)

You create and manage keys in Cloud KMS. Control rotation schedule, access policies, and lifecycle. Destroying the key makes data **permanently inaccessible**. Required for many compliance frameworks. **Autokey** simplifies CMEK provisioning.

CSEK (Customer-Supplied)

You provide the AES-256 key with each API call. Google never stores your key persistently. If you lose the key, data is **irrecoverable**. Only for GCS objects and Compute Engine disks. Highest control, highest risk.

Key Hierarchy Summary

Default: Google manages everything. **CMEK:** You manage KEK, Google manages DEK. **CSEK:** You supply KEK externally per request. **EKM:** KEK hosted in external key manager (your infrastructure).

Exam Tip: "Control encryption keys but not manage infrastructure" → CMEK.

"Keys never stored by Google" → CSEK.

"Key held in on-prem HSM" → EKM.

"Compliance requires customer key control" → CMEK at minimum.

Hosting Keys on a Hardware Security Module (HSM)

- Cloud HSM is GCP's fully managed Hardware Security Module (HSM) service
- FIPS 140-2 Level 3 certified HSM cluster
- You can choose to have KMS keys backed by HSM (on creation)

```
gcloud kms keys create key \  
  --keyring key-ring \  
  --location location \  
  --purpose "encryption" \  
  --protection-level "hsm"
```

Data Security in Transit

Google encrypts all data in transit between its data centers. For user-to-Google traffic, **TLS** is used. For service-to-service within Google, **ALTS** provides mutual authentication and encryption.

Transit Encryption Layers:

Internet → GFE: TLS 1.2+ (terminated at Google Front End)

GFE → Backend: ALTS (Application Layer Transport Security)

Between data centers: Automatic encryption at network layer

VM-to-VM same VPC: Encrypted if across hosts (BoringCrypto)

TLS / mTLS

TLS: Server proves identity to client (standard HTTPS). **mTLS:** Both client and server authenticate with certificates. Use mTLS for service-to-service in zero-trust architectures.

ALTS

Google's internal mTLS replacement. Provides mutual authentication, integrity, and encryption for all inter-service traffic. Uses service identities, not hostnames. Not directly configurable — automatic for Google services.

Certificate Manager

- Acquire and manage SSL certificates for global external HTTP(S) load balancers
- Automatically renew certificates
- You can also manually upload certificates from your/third-party CAs

Secret Manager

Secret Manager stores API keys, passwords, certificates, and other sensitive data. Secrets are **versioned**, **auditable**, and accessible via API, CLI, or client libraries. Supports CMEK encryption and automatic rotation.

Core Concepts

Secret: Named container with metadata and IAM policy. **Secret version:** Immutable payload (the actual secret data). Versions can be enabled, disabled, or destroyed. Access latest version or pin to specific version number.

IAM Roles

secretAccessor: Read secret versions (most workloads). **secretAdmin:** Full management. Grant at **individual secret** level for least privilege. does NOT include secret access.

Rotation

Configure rotation schedule with Pub/Sub topic. Trigger Cloud Function to generate new secret version and update external systems. Manual rotation: add new version, pin apps to latest or update version reference.

Service Integrations

GKE: Secrets Store CSI driver mounts secrets as volumes (Workload Identity required). Secret sync to K8s secrets (preview). **Cloud Run / Functions:** Mount as env vars or volumes. **Cloud Build:** Inject during build steps.

Exam Tip: "Store DB password securely" → Secret Manager.
"Data residency for secrets" → Regional or user-managed replication.
Editor role cannot access secrets.

Sensitive Data Protection

Sensitive Data Protection (formerly Cloud DLP) discovers, classifies, and protects sensitive data across GCS, BigQuery, Datastore, and custom sources. Uses 150+ built-in **infoTypes** (SSN, credit card, email, etc.) for detection.



Inspection Templates

Define which infoTypes to scan for and sensitivity thresholds. Reusable across inspection jobs. Custom infoTypes via regex, dictionaries, or stored custom detectors.

De-identification Templates

Define transformation rules per infoType. Apply different techniques to different data types in a single pass. Combine with KMS keys for reversible tokenization. Use for ML training data preparation.

Integration Points

Dataflow: Streaming de-identification pipeline.

BigQuery: Inspect query results in real time.

Cloud Storage: Create de-identified copies of entire buckets.

Data Profiling

Automated discovery across org, folders, or projects. Scans BigQuery tables and GCS buckets. Generates sensitivity and data risk scores. Integrates with SCC for findings.

HIPAA

- There is no recognized certification for HIPAA compliance, only rules
 - Privacy Rule
 - Security Rule
 - Enforcement Rule
- Customers that are subject to HIPAA must review and accept Google's **Business Associate Agreement (BAA)**
- The BAA covers the majority, but not all GCP services

HIPAA: Customer Responsibilities

- Execute a **Google Cloud BAA**
- Ensure that Google Cloud products not explicitly covered by the BAA aren't used when working with Protected Health Information (PHI)
- Ensure that applications are properly configured and secured according to HIPAA requirements

HIPAA: Customer Responsibilities

Technical considerations:

- Object Versioning on Cloud Storage
- Audit log exports
- Encrypt any PHI, security credentials, or other sensitive data
- Do not store PHI in metadata, including metric labels, VM labels, and GKE resource annotations
- Enforce the principle of least privilege and apply security best practices

GCP Industry Certifications

- **Compliance resource center** is the go-to place for compliance resources and offerings
- **Compliance Reports Manager** is the go-to place for downloading reports

Compliance offerings

To help you with compliance and reporting, we share information, best practices, and easy access to documentation. Our products regularly undergo independent verification of security, privacy, and compliance controls, achieving certifications against global standards to earn your trust. We're constantly working to expand our coverage.

This site contains information about Google's certifications and compliance standards it satisfies as well as general information about certain region or sector-specific regulations.

Filter by: Regions ▾ Industries ▾ Focus area ▾



ISO 9001:2015

[Learn more](#)



ISO/IEC 27001

[Learn more](#)



ISO/IEC



ISO/IEC

Common Use Case: Export for Compliance

- 1) Create a Cloud Storage bucket
- 2) (Optional) Configure object lifecycle management
- 3) Configure sink for Cloud Audit logs

Securing Vertex AI

IAM for Vertex AI

aiplatform.admin: Full access to all Vertex resources. **aiplatform.user:** Create/run training jobs, deploy models. Most resources are project-level — all authorized users share access to models in the same project. Endpoint-level IAM available for fine-grained control.

Network Security

VPC-SC perimeter: Wrap Vertex AI services to prevent data exfiltration. **Private endpoints:** Deploy prediction endpoints on private IP (VPC Network Peering). **VPC Peering:** Training and notebook VMs in your VPC.

CMEK for AI

Encrypt training data, models, endpoints, metadata, and pipeline artifacts with CMEK. Key must be in the **same region** as the resource. AutoML and custom training both support CMEK. Notebooks support CMEK for disk encryption.

Data Governance

Sensitive Data Protection: De-identify PII before training.

Vertex AI Metadata: Track data lineage and provenance across pipeline runs.

Dataset IAM: Control who can access training datasets.



Exam Tip: "Prevent model exfiltration" → VPC-SC perimeter around Vertex AI.
"Encrypt ML training data with own keys" → CMEK.
"De-identify PII before training" → Sensitive Data Protection pipeline.
"Private model endpoint" → Private endpoint with VPC peering.

Questions Breakdown

Your company wants to enforce locality requirements by limiting the physical locations where cloud resources can be created.

What should you do?

- A. Configure an organization policy with a constraint on resources location
- B. Configure an organization policy with a constraint on resource service usage
- C. Configure the organization's resource quotas and set all quotas to zero under the disallowed regions
- D. Ensure all IAM roles that create resources include a condition specifying Google Cloud regions

Questions Breakdown

Your company wants to enforce locality requirements by limiting the physical locations where cloud resources can be created.

What should you do?

- A.** Configure an organization policy with a constraint on resources location
- B.** Configure an organization policy with a constraint on resource service usage
- C.** Configure the organization's resource quotas and set all quotas to zero under the disallowed regions
- D.** Ensure all IAM roles that create resources include a condition specifying Google Cloud regions

Questions Breakdown

Your company's security requirements dictate that all data must be encrypted at rest with encryption keys that can be manually rotated by the security team when needed.

How should you configure the storage services on GCP?

- A. Ensure that all storage services are created using customer-supplied encryption keys (CSEKs)
- B. Ensure that only storage services with Google-managed encryption keys (GMEKs) are used
- C. Generate an AES-256 key using a third-party tool. Store the key in a storage bucket that the security team has access to. Use this key to encrypt all data
- D. Ensure that all storage services are created using customer-managed encryption keys (CMEKs)

Questions Breakdown

Your company's security requirements dictate that all data must be encrypted at rest with encryption keys that can be manually rotated by the security team when needed.

How should you configure the storage services on GCP?

- A. Ensure that all storage services are created using customer-supplied encryption keys (CSEKs)
- B. Ensure that only storage services with Google-managed encryption keys (GMEKs) are used
- C. Generate an AES-256 key using a third-party tool. Store the key in a storage bucket that the security team has access to. Use this key to encrypt all data
- D. Ensure that all storage services are created using customer-managed encryption keys (CMEKs)

Questions Breakdown

Your ML team is building a customer churn prediction model on Vertex AI. The training dataset contains customer PII including names, emails, and phone numbers. The model must be deployed to a private endpoint accessible only within the VPC, and the training data must be deidentified before model training.

What should you do?

- A.** Use BigQuery column-level security to mask PII columns, train the model on masked data, and deploy to a public endpoint with IAM
- B.** Use Sensitive Data Protection to de-identify PII with format-preserving encryption, train the model, and deploy to a private endpoint with VPC peering
- C.** Manually redact PII from the training CSV, upload to GCS with CMEK, and deploy behind Cloud Armor
- D.** Use Dataflow with custom PII detection to anonymize data, train the model, and deploy with IAP in front of the endpoint

Questions Breakdown

Your ML team is building a customer churn prediction model on Vertex AI. The training dataset contains customer PII including names, emails, and phone numbers. The model must be deployed to a private endpoint accessible only within the VPC, and the training data must be deidentified before model training.

What should you do?

- A. Use BigQuery column-level security to mask PII columns, train the model on masked data, and deploy to a public endpoint with IAM
- B. Use Sensitive Data Protection to de-identify PII with format-preserving encryption, train the model, and deploy to a private endpoint with VPC peering**
- C. Manually redact PII from the training CSV, upload to GCS with CMEK, and deploy behind Cloud Armor
- D. Use Dataflow with custom PII detection to anonymize data, train the model, and deploy with IAP in front of the endpoint

Questions Breakdown

You're managing a secured Google Cloud project with several Compute Engine instances. No instances in the project can have a public IP address and there is no hybrid connectivity between Google Cloud and your company's office. You need to connect via SSH into a specific machine for troubleshooting.

What should you do?

- A.** Delete any organization policy constraints in place that prevent the assignment of external IP addresses to Compute Engine instances. Assign an external IP address to the instance during troubleshooting.
- B.** Configure Cloud NAT on the instance's subnet and a VPC Firewall rule allowing SSH connections from your IP address. Connect via SSH using Cloud NAT's IP address.
- C.** Configure Identity-Aware Proxy (IAP) with TCP forwarding and a VPC Firewall rule allowing SSH connections from the proxy. Connect to the instance using gcloud.
- D.** Create a bastion host in the same VPC network and a VPC Firewall rule allowing SSH connections from the bastion host. Connect to the instance via the bastion host's SSH client.

Questions Breakdown

You're managing a secured Google Cloud project with several Compute Engine instances. No instances in the project can have a public IP address and there is no hybrid connectivity between Google Cloud and your company's office. You need to connect via SSH into a specific machine for troubleshooting.

What should you do?

- A.** Delete any organization policy constraints in place that prevent the assignment of external IP addresses to Compute Engine instances. Assign an external IP address to the instance during troubleshooting.
- B.** Configure Cloud NAT on the instance's subnet and a VPC Firewall rule allowing SSH connections from your IP address. Connect via SSH using Cloud NAT's IP address.
- C.** Configure Identity-Aware Proxy (IAP) with TCP forwarding and a VPC Firewall rule allowing SSH connections from the proxy. Connect to the instance using `gcloud`.
- D.** Create a bastion host in the same VPC network and a VPC Firewall rule allowing SSH connections from the bastion host. Connect to the instance via the bastion host's SSH client.

Segment 7: Deployment and operational best practices



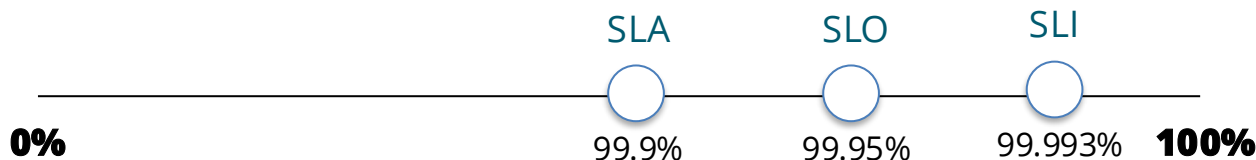
Objectives

- Google Cloud Observability (monitoring and logging, profiling, alerting)
- Troubleshooting and incident management
- Interacting with GCP programmatically
- Automating infrastructure and application deployments
- Testing and validating deployments
- API management best practices
- Gemini Cloud Assist

SLIs, SLOs, and SLAs

- **SLI**: Quantifiable measure of service reliability
- **SLO**: Reliability target for an SLI
- **SLA**: Agreed-upon reliability target as a commitment to paying customers

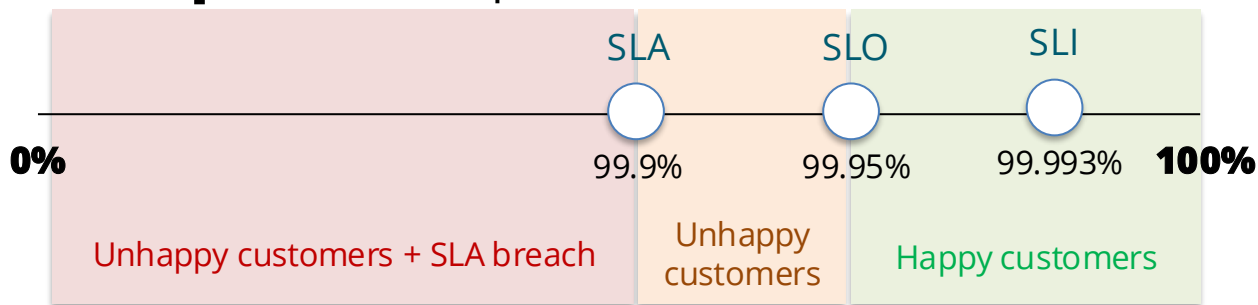
Example: service uptime



SLIs, SLOs, and SLAs

- **SLI**: Quantifiable measure of service reliability
- **SLO**: Reliability target for an SLI
- **SLA**: Agreed-upon reliability target as a commitment to paying customers

Example: service uptime



Google Cloud Operations Suite

Monitoring

Logging
and Error
Reporting

Managed
Service for
Prometheus

Application Performance Monitoring

Trace

Profiler

Google Cloud Observability

Google Cloud Observability (formerly Stackdriver) is an integrated suite for monitoring, logging, tracing, and debugging cloud applications. It provides visibility into performance, uptime, and overall health across GCP and hybrid environments.

Cloud Monitoring

Collect metrics, create dashboards, configure alerting policies. Auto-collects predefined metrics from GCP services at no cost.

Cloud Logging

Ingest, store, search, and analyze log data. Log Router directs entries to sinks (Cloud Storage, BigQuery, Pub/Sub).

Cloud Trace

Distributed tracing for latency analysis. Find performance bottlenecks across microservices with trace spans.

Cloud Profiler

Continuous CPU and heap profiling with minimal overhead. Analyze production performance in Go, Java, Node.js, Python.

Error Reporting

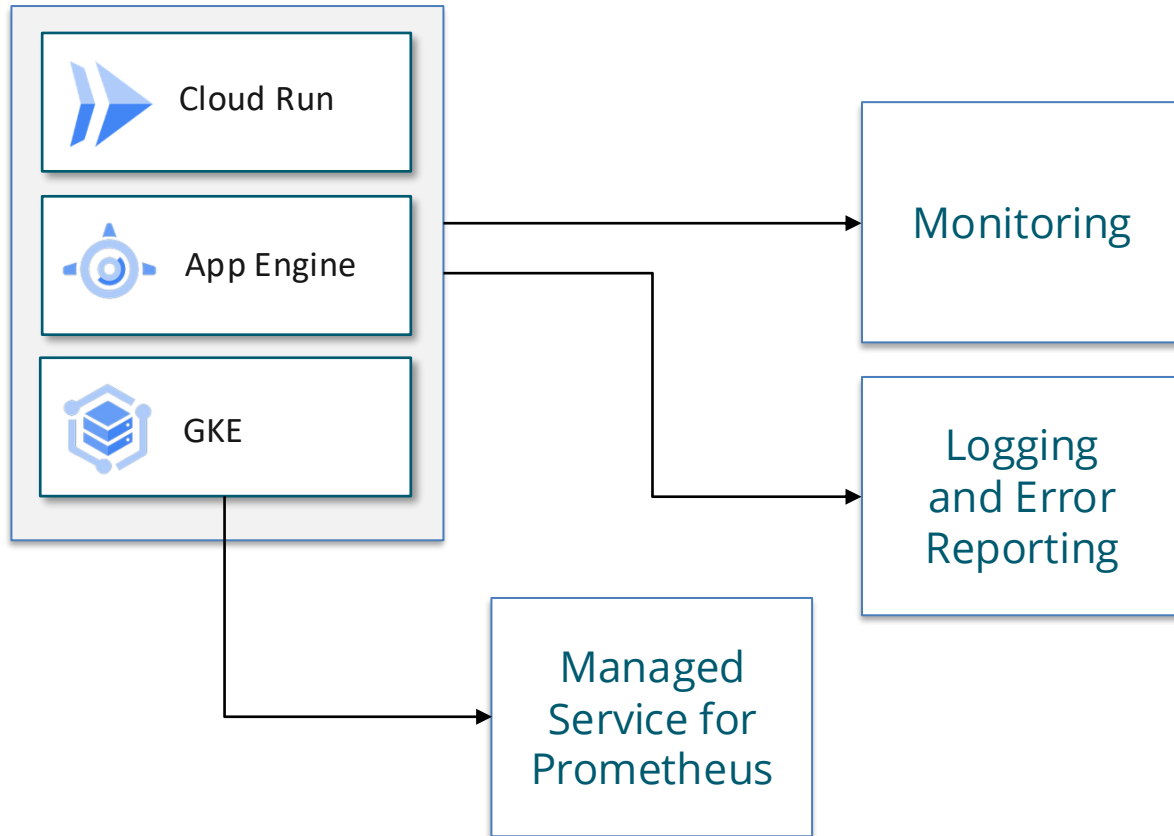
Aggregate and group error events by root cause. Supports Cloud Run, GKE, App Engine, Cloud Functions, Compute Engine.

Uptime Checks

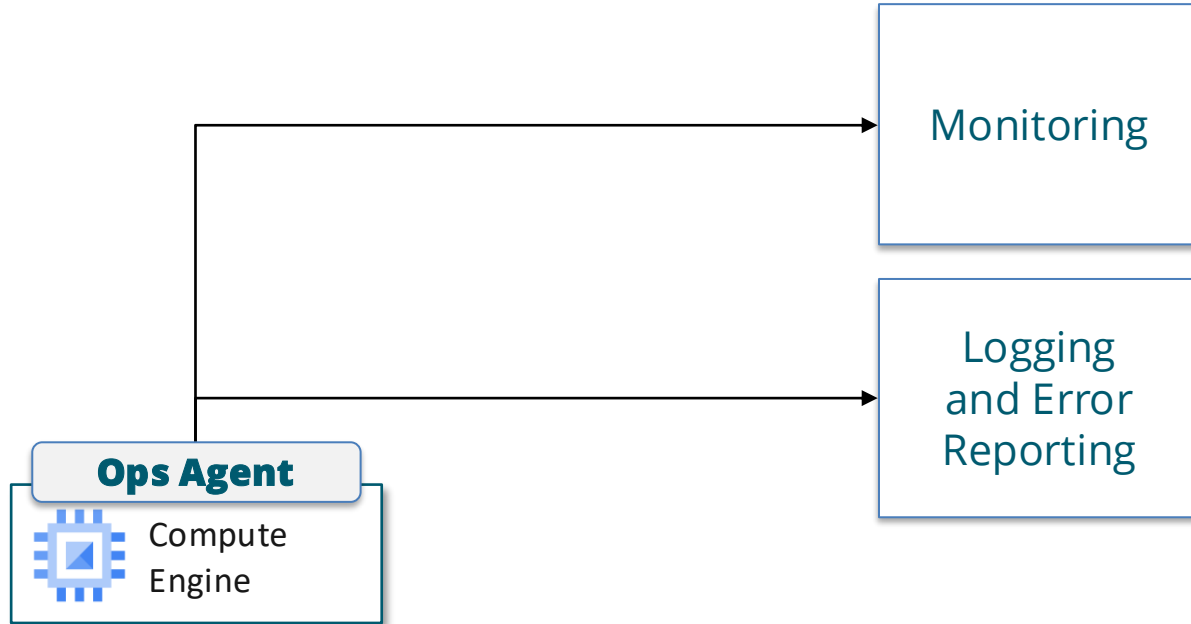
Periodically probe HTTP(S), TCP endpoints from global locations. Trigger alerts when checks fail.

Exam Tip: Cloud Monitoring, Logging, Trace, and Error Reporting share a common navigation pane for integrated troubleshooting workflows.

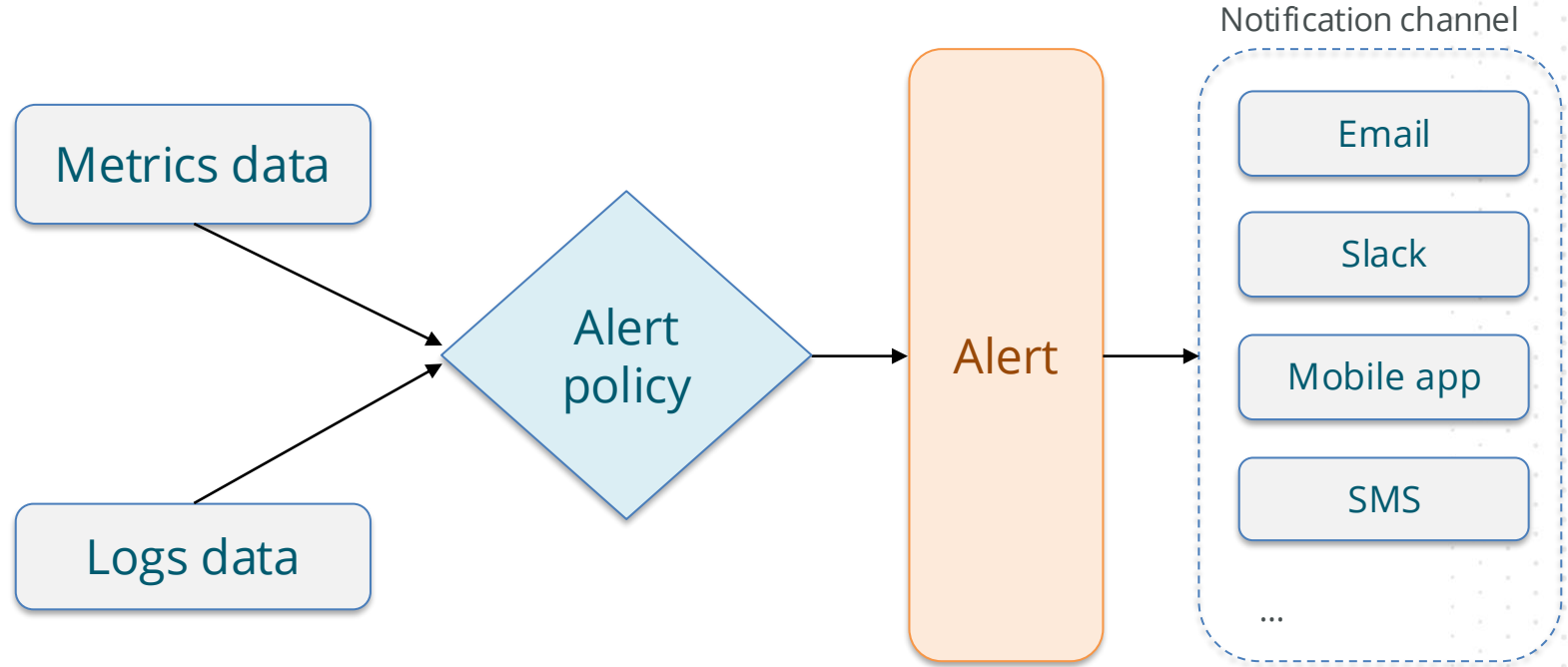
Metrics and Logs Collection



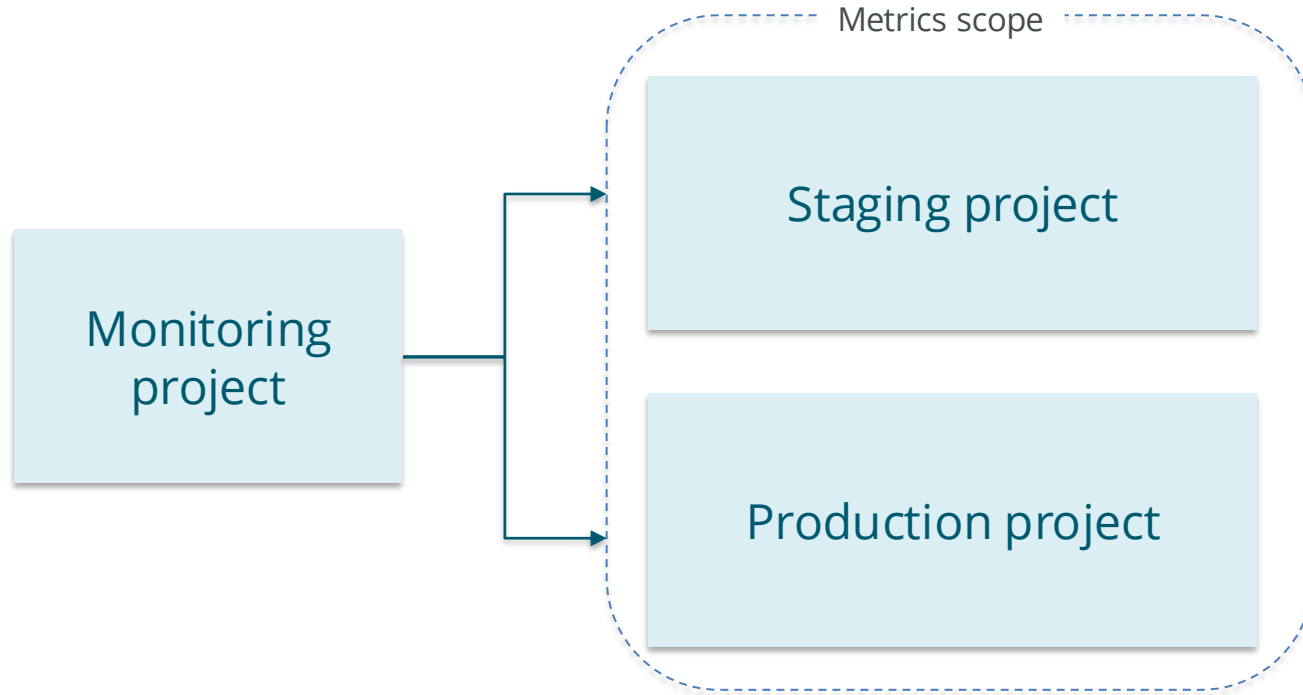
Metrics and Logs Collection



Metrics and Alerts

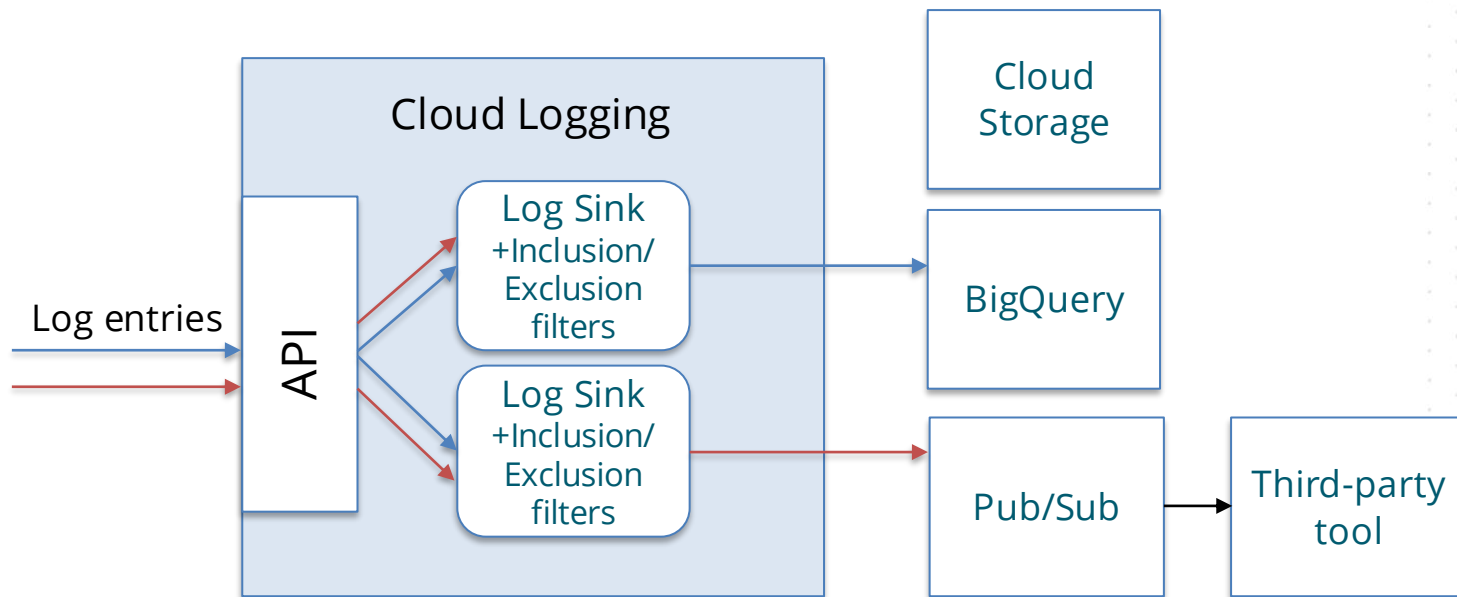


Metric Scopes



Cloud Logging: Sinks

Routing behavior is controlled with inclusion filters and exclusion filters



Cloud Trace

- Distributed tracing system that collects **latency** data from applications in near real-time
- Helps answer questions :
 - *What is the overall latency of requests?*
 - *How long does it take to handle a given request?*
 - *Why is it taking so long to handle a request?*
 - *Why do some requests take longer than others?*

Cloud Trace and Cloud Profiler

Cloud Trace provides distributed tracing for latency bottlenecks. Cloud Profiler profiles CPU and memory with minimal overhead.

Cloud Trace

- Distributed tracing across microservices
- Trace Explorer: heatmaps, span grouping
- Trace Scopes: project-level search configs
- OpenTelemetry instrumentation support

When to Use Trace

- Identify slow service calls in request path
- Find latency bottlenecks in microservices
- Correlate traces with logs and errors

Cloud Profiler

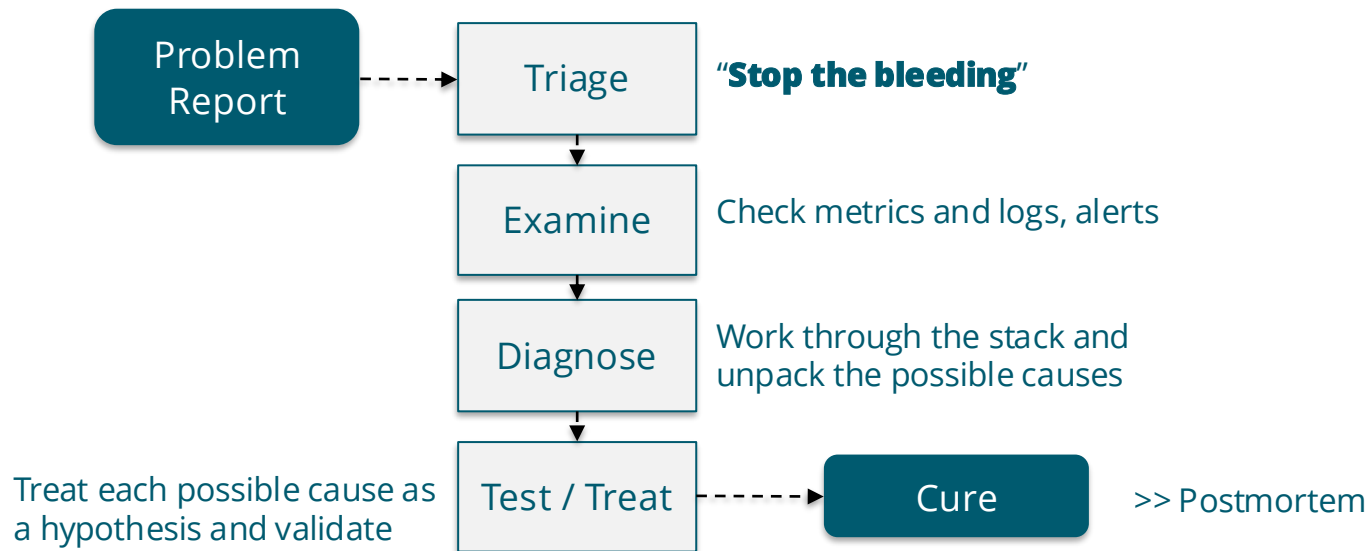
- Continuous CPU and heap profiling
- Minimal overhead (safe for production)
- Flame graphs show call hierarchy
- Supports Go, Java, Node.js, Python
- No code changes needed (agent-based)

When to Use Profiler

- Find CPU-intensive functions
- Identify memory leaks and hotspots
- Compare performance before/after changes

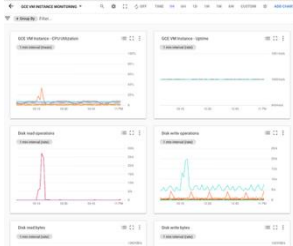
Troubleshooting Best Practices

- *Blameless post-mortems* culture
- Follow the **Site Reliability Engineering (SRE)** troubleshooting process



Developing a Structured Incident Response

Monitoring Dashboard



Alerting Strategy



Response Strategy



Monitoring Dashboard

SLI/SLO Dashboard

("Black box" view)

Are we meeting our
promises to our
users?

What is our error
budget consumption?

Service Health Dashboard

("Clear box" view)

Why is the service
unhealthy?

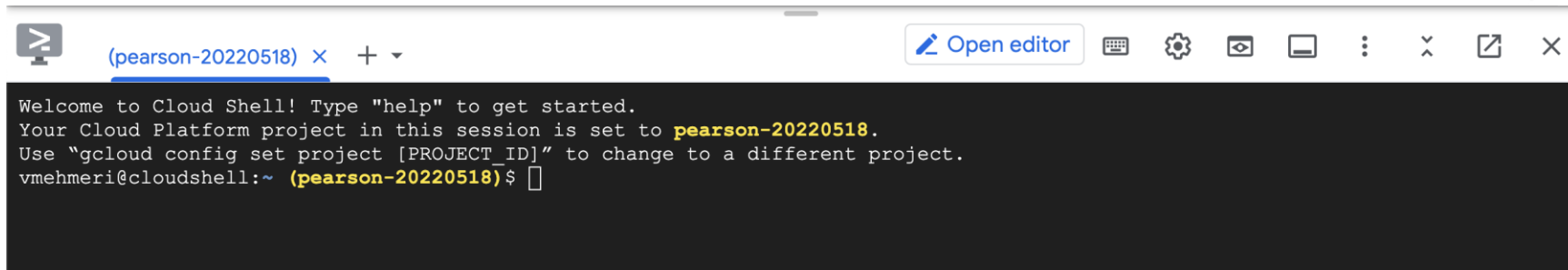
SRE "golden signals":
latency, traffic, errors,
and saturation

Incident Response

- Have a documented and well-exercised incident management plan
- Assign clear ownership (“Incident Commander”)
- Open a communication hub (e.g. Slack channel)
- “Stop the bleeding” first
- Schedule a blameless postmortem

Cloud Shell

- Browser-based, interactive shell environment
- Tools pre-installed (*gcloud*, *gsutil*, *bq*, etc.)
- Built-in code editor
- Runs on a shared VM (convenient, but no SLA)



```

Welcome to Cloud Shell! Type "help" to get started.
Your Cloud Platform project in this session is set to pearson-20220518.
Use "gcloud config set project [PROJECT_ID]" to change to a different project.
vmehmeri@cloudshell:~ (pearson-20220518)$
```

Google Cloud CLI (gcloud)

- Command-line interface (*gcloud*)
- Set of tools to create and manage GCP resources

Basic Syntax - gcloud

gcloud + component + entity + operation + positional args + flags

```
$ gcloud compute instances attach-disk INSTANCE_NAME --disk=DISK --project=PROJECT_ID
```

Diagram illustrating the components of the command:

- component**: `compute`
- entity**: `instances`
- operation**: `attach-disk`
- pos. arg**: `INSTANCE_NAME`
- flags**: `--disk=DISK --project=PROJECT_ID`

Cloud Storage Utility (gsutil)

- You get *gsutil* as part of the Google Cloud CLI
- Specific to Cloud Storage for running a wide range of bucket and object management tasks
 - Creating and deleting buckets
 - Uploading, downloading, and deleting objects
 - Listing buckets and objects
 - Editing object and bucket ACLs
 - Etc.

BigQuery Command-Line Tool (bq)

- Can be used to interact with BigQuery service
 - Create datasets and tables
 - Create views
 - List datasets, tables, and views
 - Load data into BigQuery
 - Run queries
 - Etc.

Code and Environment Management

- All code in a code repository
- Explicitly declare dependencies (including versions) and install them using a dependency manager
- Separate configuration settings from code

**Source
repository**
(code only)

**Dependency
management**

**Decoupled
configuration**

Code and Environment Management

Example

**Cloud Source
Repositories**

NodeJS

**Node Package
Manager (NPM)**

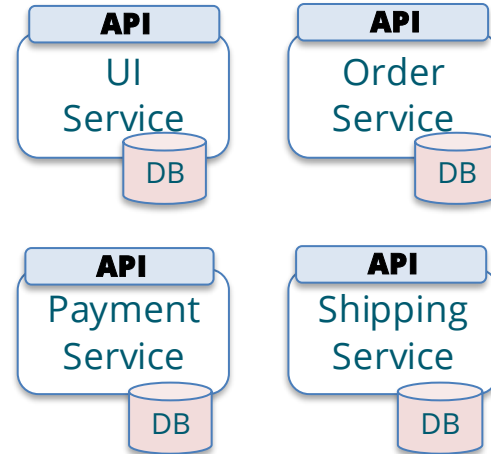
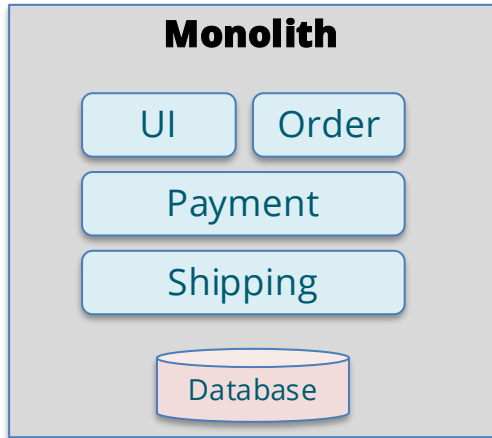
package.json

**Environment
variables**

CONFIG=val

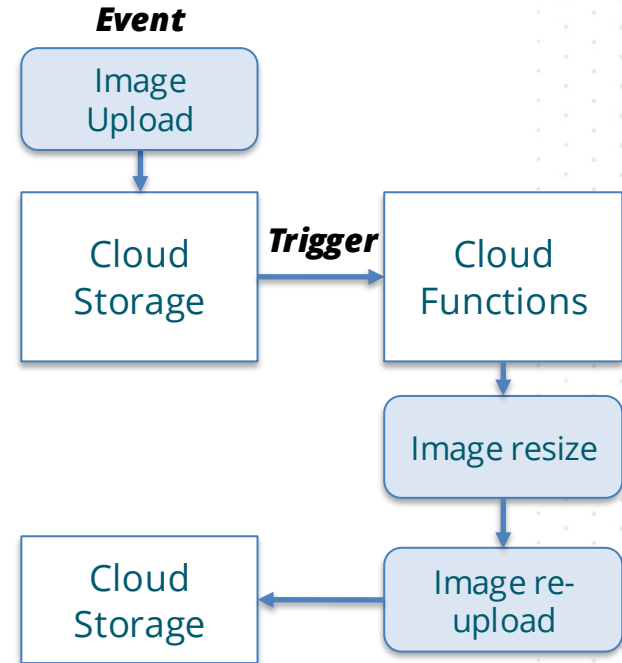
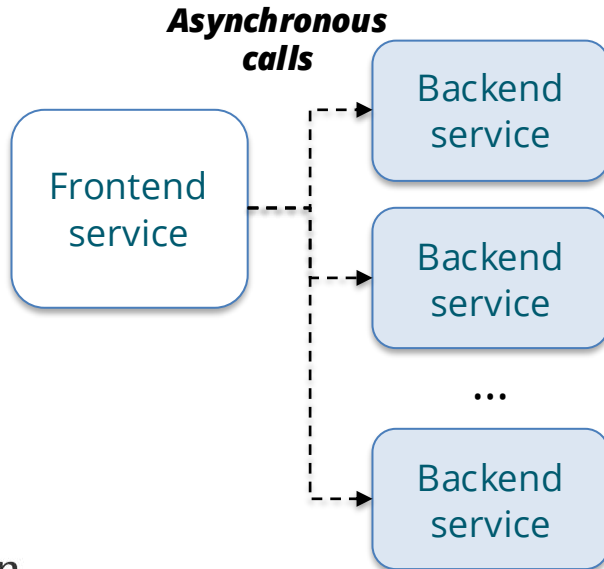
Application Design: Microservices

Favor **microservices** over monolithic designs if the goal is scaling



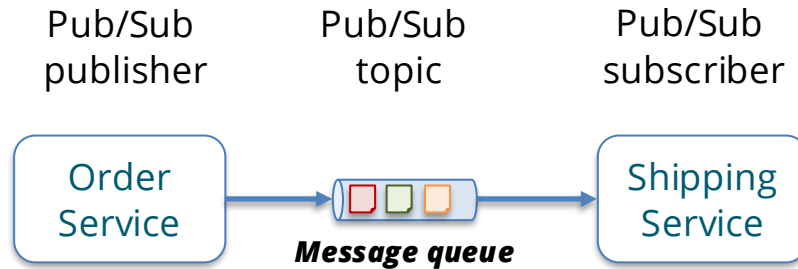
Application Design: Asynchronous, Event-Driven

- Favor **asynchronous** operations when possible
- Use event-driven processing



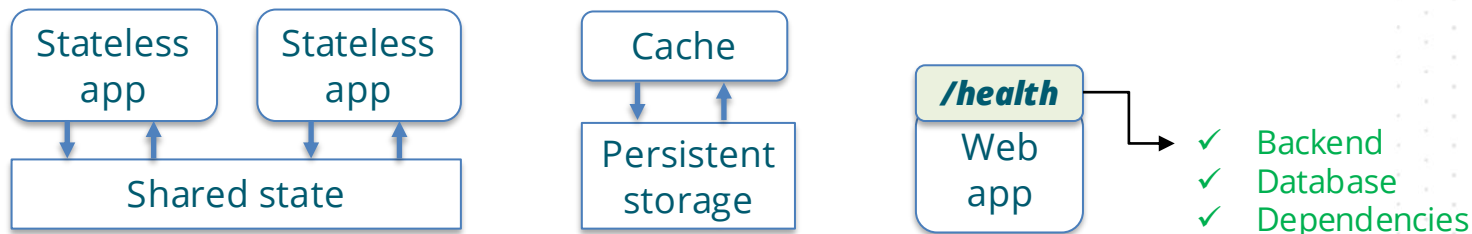
Application Design: Loose Coupling

Design for **loose coupling**

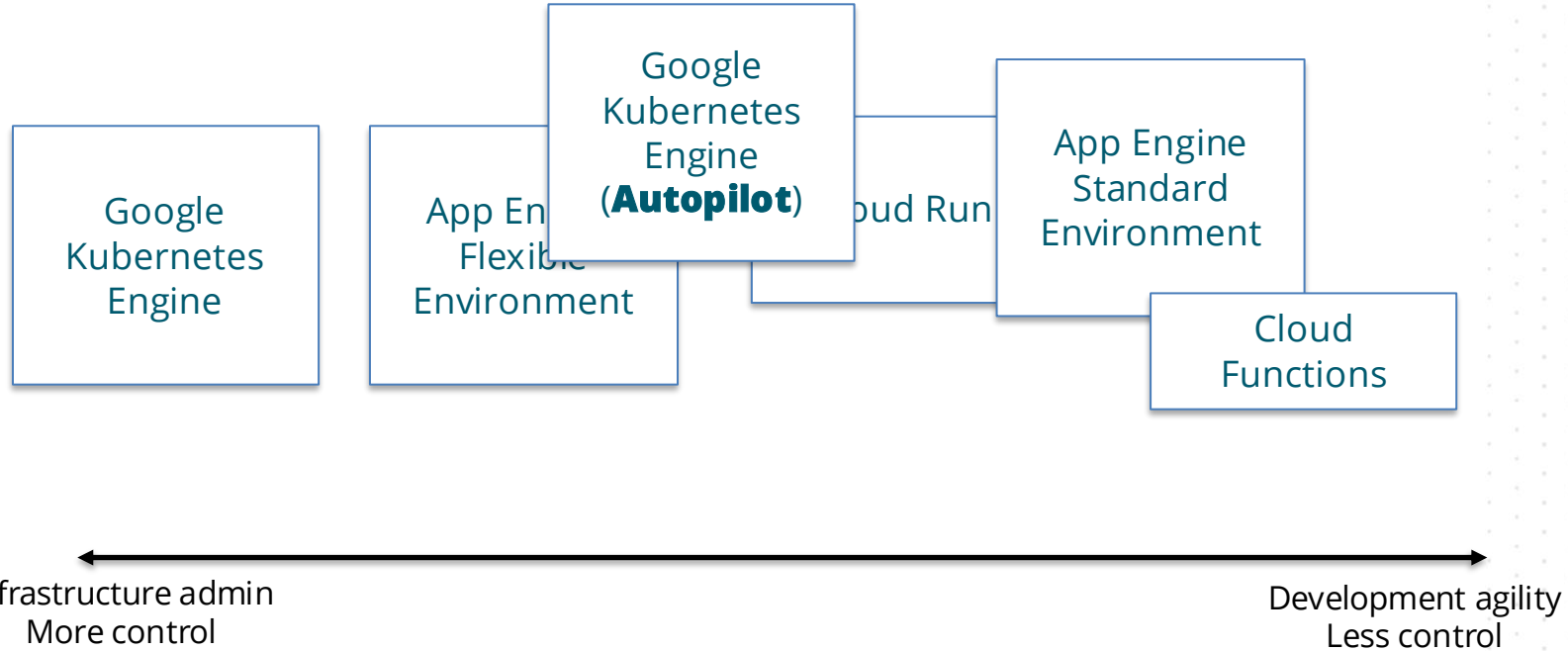


Building Scalable and Reliable Applications

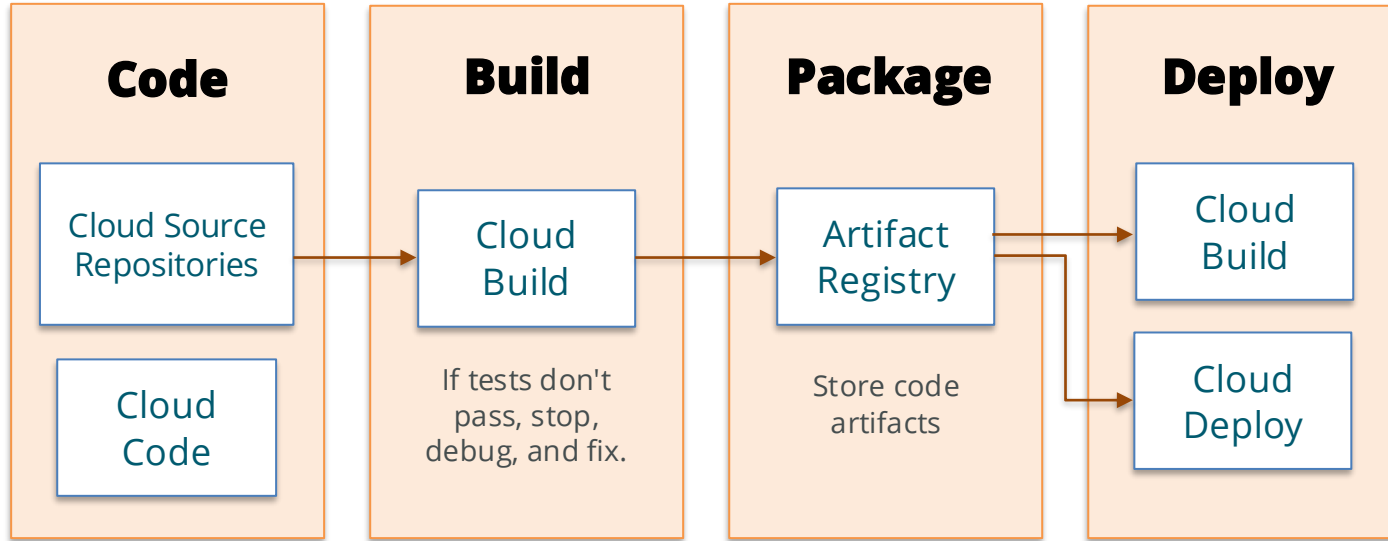
- Implement stateless components
- Cache content (Memorystore, Cloud CDN)
- Implement health-check endpoints (*/health*)



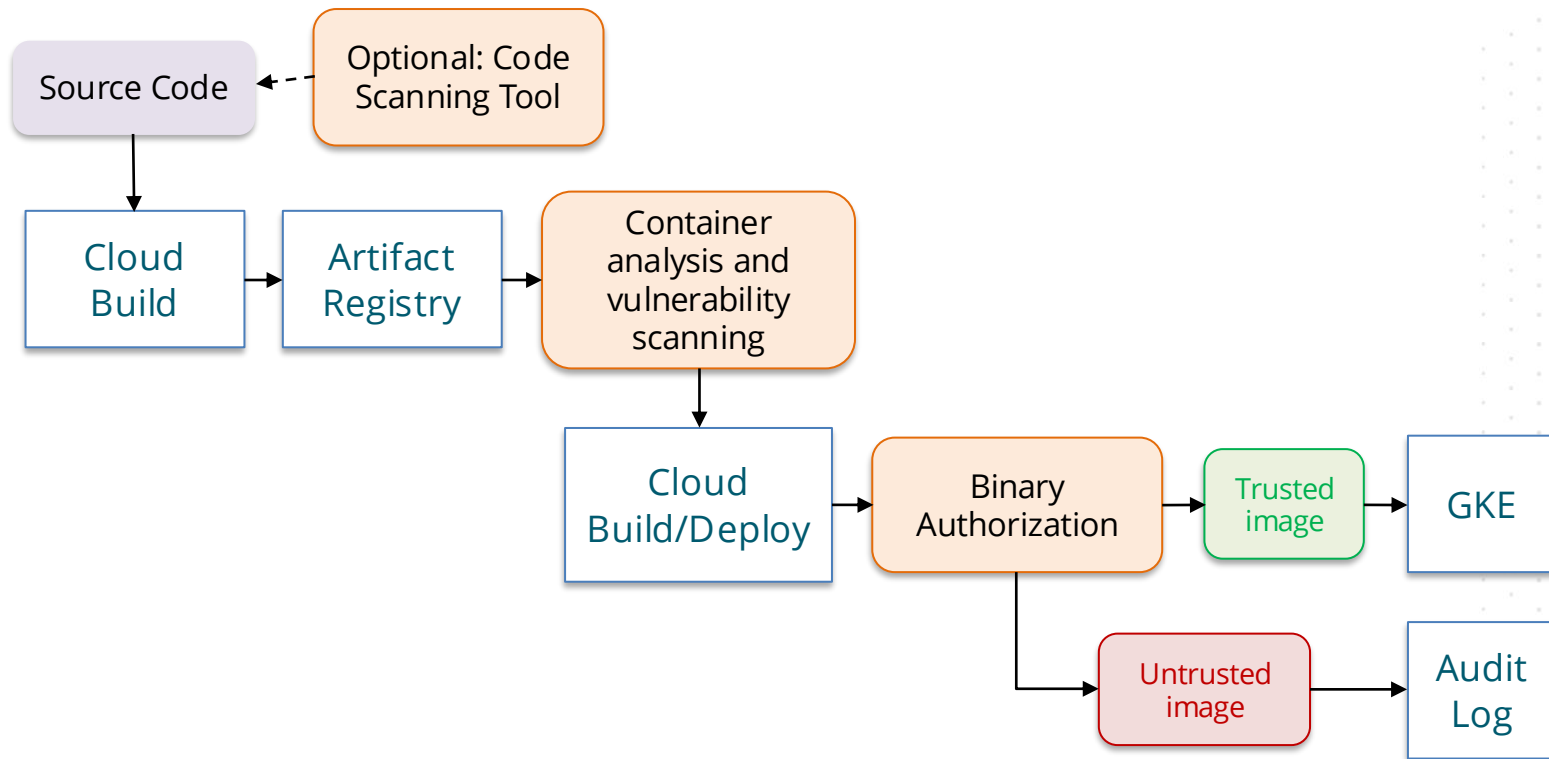
Application Development on GCP



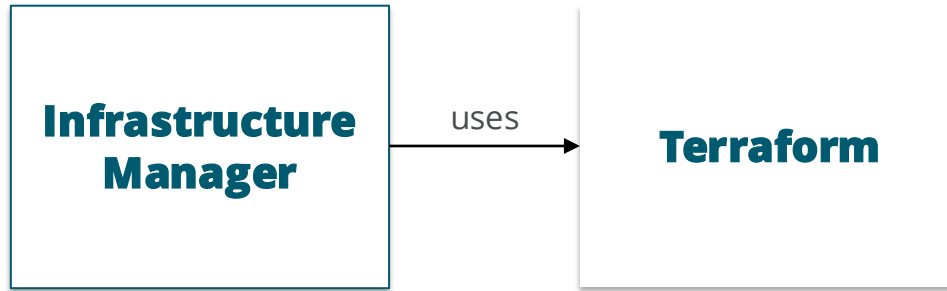
CI/CD on Google Cloud



DevSecOps with GKE



Define Infrastructure as Code



Other Ways to Automate Things

**Cloud
Scheduler**

Workflows

**Cloud
Composer**

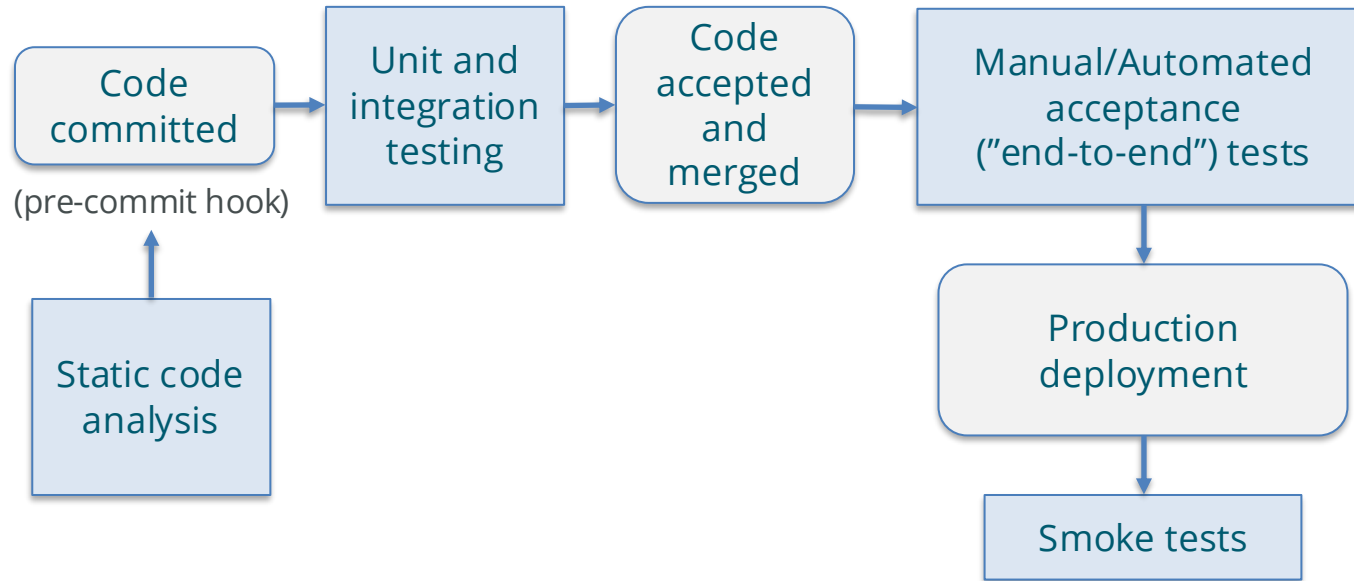
(Apache Airflow)

Deployment Automation Best Practices

- Use the same deployment process for every environment
- Use the same packages for every environment (**tip: containers**)
- Make it possible to recreate the state of any environment from version control (**tip: infrastructure-as-code**)

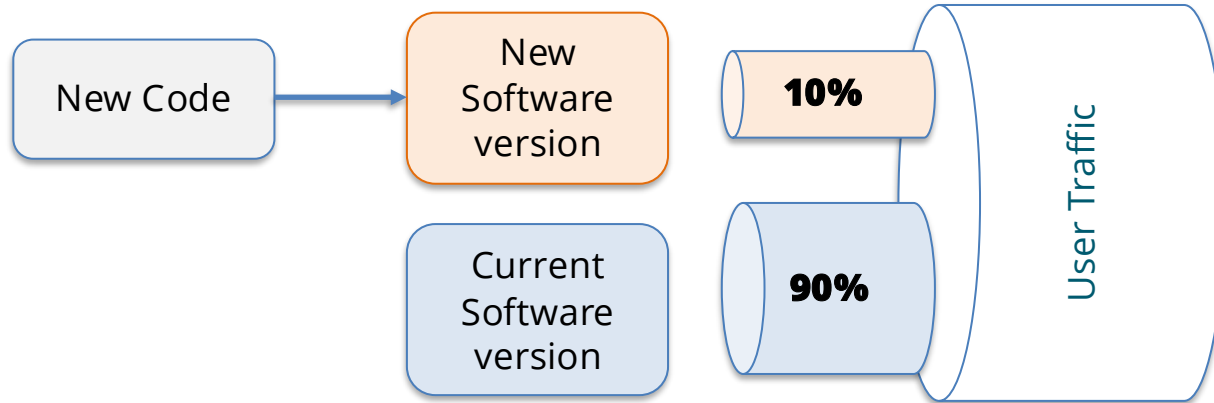
When and How to Test

Test **early** and **automate**



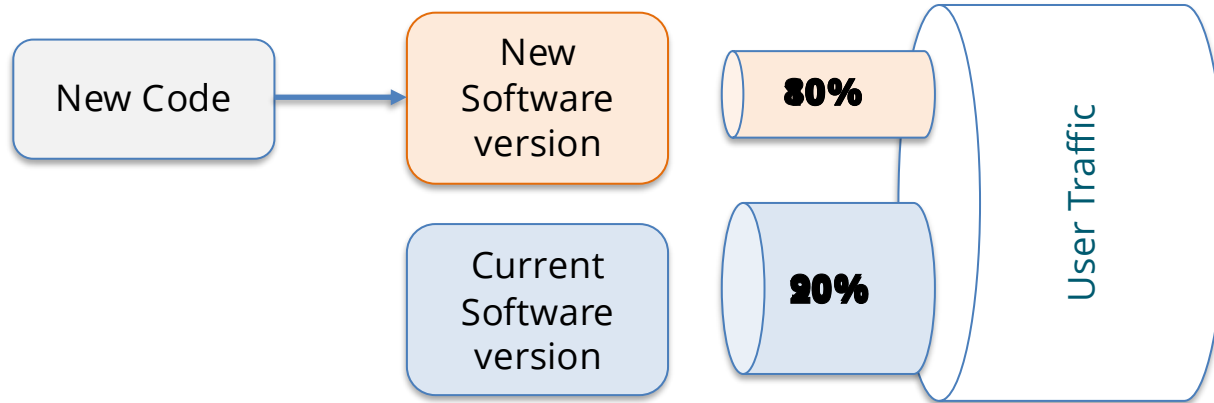
Canary Testing

Roll out new versions incrementally



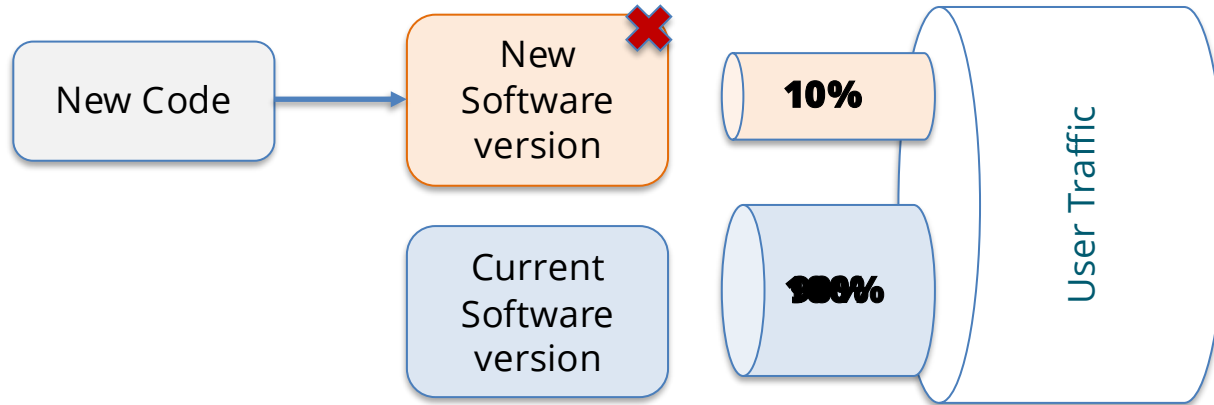
Canary Testing

Roll out new versions incrementally



Canary Testing

Roll back if there's an issue



Only 10% of users affected

Canary Testing on GCP

- Cloud Run

```
gcloud run services update-traffic SERVICE_NAME \
--to-revisions=${PROD}=90,${CANARY}=10 \
--platform managed
```

- GKE

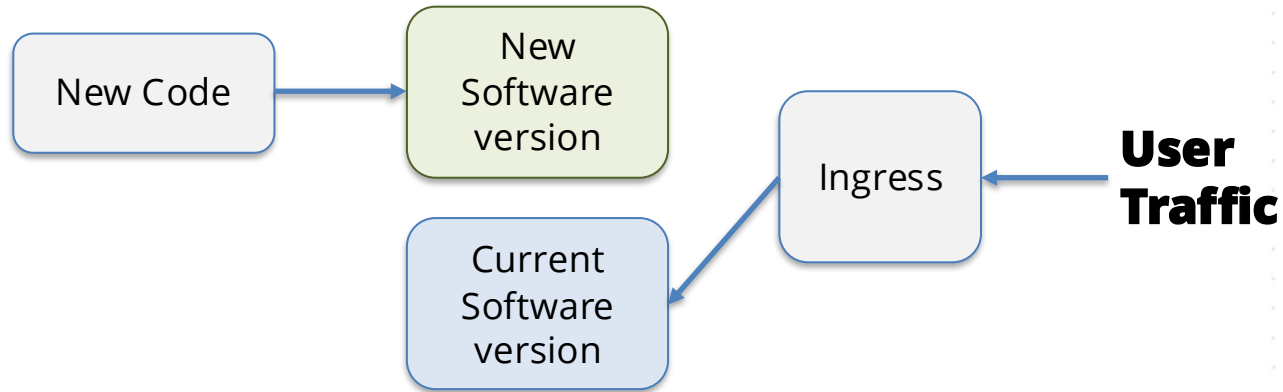
- Expose canary and current deployments under the same service
- Use the number of replicas to dictate traffic percentage
- Alternatively, use **Istio** / **Anthos Service Mesh**

- Cloud Deploy

- Automates Canary and Blue Green deployments into Cloud Run / GKE

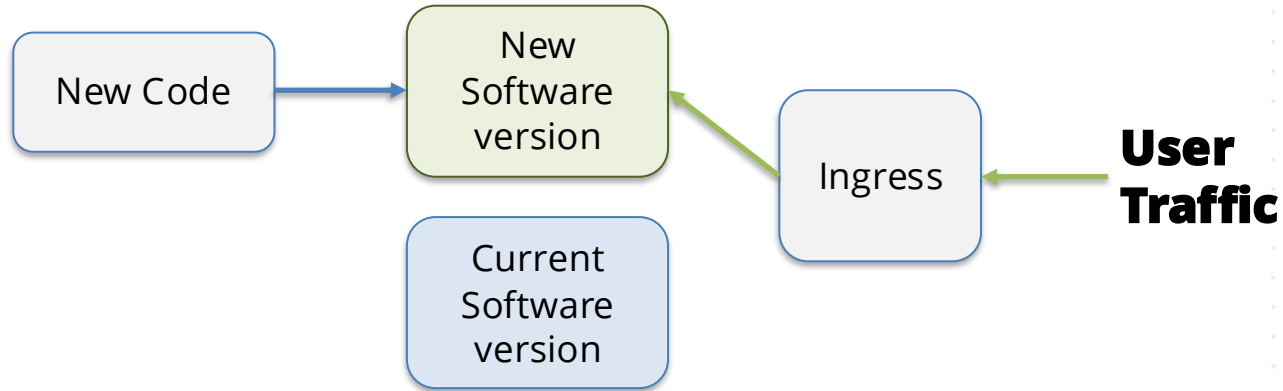
Blue/Green Deployment and Testing

Roll out new version alongside current, then switch



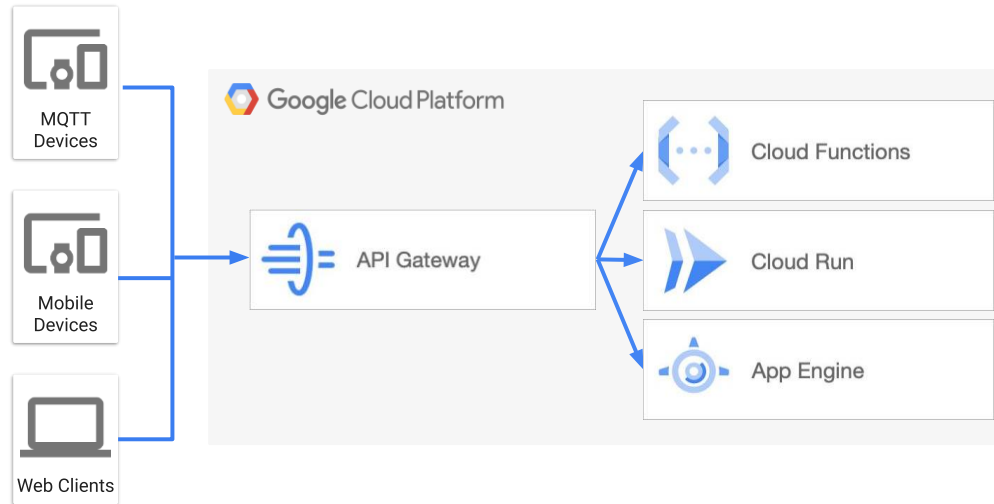
Blue/Green Deployment and Testing

Roll out new version alongside current, then switch



API Gateway

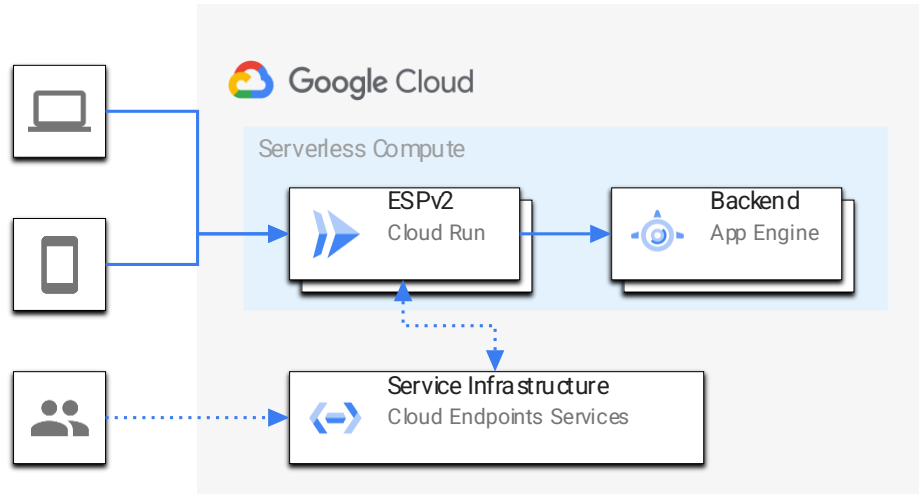
GCP's **API Gateway** is a fully managed gateway for serverless backends.



No developer portal or advanced features

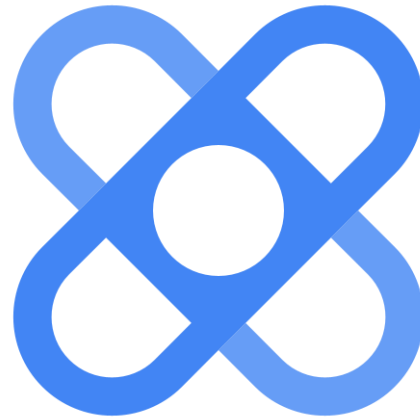
Cloud Endpoints

- You install Google Cloud's Extensible Service Proxy (ESPV2)
- You have full control over the API gateway
- GCP-based backends



Apigee

- Apigee is a fully featured API management platform
- Can proxy APIs outside of Google Cloud
- "Enterprise" feature set:
 - Advanced security
 - Rate-limiting
 - Quotas
 - Monetization
 - Deep analytics
 - Customizable portals



Questions Breakdown

Your company runs a microservices application on GKE. The SRE team needs to identify latency bottlenecks in production when users report slow responses. The services communicate via gRPC. You need to find which specific service calls are causing delays with minimal code changes.

Which Google Cloud service should you use?

- A. Cloud Profiler to analyze CPU usage across services
- B. Cloud Trace to visualize request latency across service calls
- C. Cloud Monitoring with custom metrics for each service
- D. Cloud Logging with log-based metrics to track request times

Questions Breakdown

Your company runs a microservices application on GKE. The SRE team needs to identify latency bottlenecks in production when users report slow responses. The services communicate via gRPC. You need to find which specific service calls are causing delays with minimal code changes.

Which Google Cloud service should you use?

- A. Cloud Profiler to analyze CPU usage across services
- B. Cloud Trace to visualize request latency across service calls
- C. Cloud Monitoring with custom metrics for each service
- D. Cloud Logging with log-based metrics to track request times

Questions Breakdown

Your team is deploying a critical application to Cloud Run. You need to gradually roll out new versions to production, starting with a small subset of traffic on the new versions. If metrics show elevated error rates during the rollout, you want automatic rollback without manual intervention.

Which deployment approach should you use?

- A. Blue-green deployment with manual traffic switching
- B. Standard rolling deployment with Cloud Monitoring alerts
- C. Cloud Run traffic splitting with manual percentage updates
- D. Cloud Deploy with canary strategy and automated rollback rules

Questions Breakdown

Your team is deploying a critical application to Cloud Run. You need to gradually roll out new versions to production, starting with a small subset of traffic on the new versions. If metrics show elevated error rates during the rollout, you want automatic rollback without manual intervention.

Which deployment approach should you use?

- A. Blue-green deployment with manual traffic switching
- B. Standard rolling deployment with Cloud Monitoring alerts
- C. Cloud Run traffic splitting with manual percentage updates
- D. Cloud Deploy with canary strategy and automated rollback rules**

Questions Breakdown

You are designing a new application that has several configuration settings that are environment-specific. Your application also needs to connect to different databases, and you want to store the database credentials securely.

What should you do?

- A. Store the environment-specific configurations in the source code, using different branches. Store the credentials in environment variables.
- B. Store the environment-specific configurations in environment variables. Store the credentials in the source code.
- C. Store the environment-specific configurations in environment variables. Store the credentials in Secret Manager.
- D. Store both the environment-specific configurations and the database credentials in environment variables.

Questions Breakdown

You are designing a new application that has several configuration settings that are environment-specific. Your application also needs to connect to different databases, and you want to store the database credentials securely.

What should you do?

- A. Store the environment-specific configurations in the source code, using different branches. Store the credentials in environment variables.
- B. Store the environment-specific configurations in environment variables. Store the credentials in the source code.
- C. Store the environment-specific configurations in environment variables. Store the credentials in Secret Manager.
- D. Store both the environment-specific configurations and the database credentials in environment variables.

Questions Breakdown

Your team is developing a new application to be deployed on a Google Kubernetes Engine (GKE) cluster. You want to have any changes pushed to the main branch on your team's GitHub repository to automatically initiate a build that will package the application and test it.

- A.** Have each developer configure and run a test suite on their workstation that builds and tests the code before committing to a remote branch.
- B.** Spin up a test environment on GCP and create a custom script that checks for changes in the main branch of the GitHub repository, runs build and test steps, and publishes validated artifacts to Artifact Registry.
- C.** Configure a Cloud Build build config that includes the required build and test steps. Configure a Cloud Build trigger for changes to the main branch.
- D.** Configure a Cloud Build config that includes the required build steps. Ensure the built-in Container Analysis functionality is enabled to run code tests.

Questions Breakdown

Your team is developing a new application to be deployed on a Google Kubernetes Engine (GKE) cluster. You want to have any changes pushed to the main branch on your team's GitHub repository to automatically initiate a build that will package the application and test it.

What should you do to enforce this process?

- A. Have each developer configure and run a test suite on their workstation that builds and tests the code before committing to a remote branch.
- B. Spin up a test environment on GCP and create a custom script that checks for changes in the main branch of the GitHub repository, runs build and test steps, and publishes validated artifacts to Artifact Registry.
- C. Configure a Cloud Build build config that includes the required build and test steps. Configure a Cloud Build trigger for changes to the main branch.
- D. Configure a Cloud Build config that includes the required build steps. Ensure the built-in Container Analysis functionality is enabled to run code tests.

Questions Breakdown

You're responsible for the security aspect of the software development life cycle (SDLC) of your team's application that runs on Google Kubernetes Engine (GKE). You want to ensure that only trusted container images are deployed to GKE clusters.

What should you do?

- A.** Use Cloud Build to define a build step that checks if the code comes from your team's source repository and marks the container as trusted.
- B.** Create a Cloud Function that is triggered each time there are code changes on the repository. Have the Cloud Function deploy the container only if all functional tests are successful.
- C.** Configure Cloud IAM so that only trusted principals have permissions to deploy containers on GKE.
- D.** Enable Binary Authorization on the GKE clusters and configure a Binary Authorization policy.

Questions Breakdown

You're responsible for the security aspect of the software development life cycle (SDLC) of your team's application that runs on Google Kubernetes Engine (GKE). You want to ensure that only trusted container images are deployed to GKE clusters.

What should you do?

- A.** Use Cloud Build to define a build step that checks if the code comes from your team's source repository and marks the container as trusted.
- B.** Create a Cloud Function that is triggered each time there are code changes on the repository. Have the Cloud Function deploy the container only if all functional tests are successful.
- C.** Configure Cloud IAM so that only trusted principals have permissions to deploy containers on GKE.
- D.** Enable Binary Authorization on the GKE clusters and configure a Binary Authorization policy.

Questions Breakdown

You need to improve the reliability and quality of software released in production by reducing the number of rollbacks and the rollback time for faulty software.

What should you do?

(Choose two.)

A. Reduce the number of database writes.

B. Re-architect the application into microservices.

C. Adopt a blue/green deployment model.

D. Automate software tests in a CI/CD pipeline.

E. Migrate from zonal to regional resources on GCP.

Questions Breakdown

You need to improve the reliability and quality of software released in production by reducing the number of rollbacks and the rollback time for faulty software.

What should you do?

(Choose two.)

A. Reduce the number of database writes.

B. Re-architect the application into microservices.

C. Adopt a blue/green deployment model.

D. Automate software tests in a CI/CD pipeline.

E. Migrate from zonal to regional resources on GCP.

Segment 8: Case studies solutioning



Objectives

- Case Study Breakdown: Altostrat Media
- Case Study Breakdown: Cymbal Retail
- Case Study Breakdown: EHR Healthcare
- Case Study Breakdown: KnightMotives Automotive

Case Study: Altostrat Media

Company Profile

Prominent digital media company with extensive audio/video content library including podcasts, interviews, news broadcasts, and documentaries.

Business Goals

- Modernize content management with Gen AI
- Deliver personalized content recommendations
- Enable natural language interactions
- Drive revenue through dynamic pricing
- Implement targeted marketing and suggestions

Solution Concept

Platform that empowers customers with:

- AI-powered personalized recommendations
- Natural language content discovery
- Seamless self-service support
- Dynamic pricing optimization
- Automated content analysis at scale

Key Constraints

- Maintain high availability for streaming
- Optimize storage costs as library grows
- Integrate AI without disrupting existing services

Case Study: Altostrat Media

Compute

GKE clusters for scalability and high availability

Storage

Cloud Storage for media library (audio, video, docs)

Analytics

BigQuery for user behavior and content consumption

Serverless

Cloud Run Functions for transcoding and metadata

Identity

Google Identity + third-party providers

AI/ML

Vertex AI for content recommendations

Legacy Systems

Some on-premises systems still handle content ingestion and archival. These require hybrid connectivity consideration.

Current Capabilities

- Video transcoding via Cloud Run Functions
- Metadata extraction pipelines
- Audience demographic analytics in BigQuery
- Content consumption pattern tracking

Case Study: Altostrat Media

Content Processing Pipeline

- **Cloud Storage** with lifecycle policies for tiered media storage
- **Transcoder API** for video format conversion
- **Cloud Run Functions** for event-driven metadata extraction
- **Pub/Sub** for decoupled processing workflows

AI-Powered Recommendations

- **Vertex AI** for recommendation models
- **BigQuery ML** for user behavior analysis
- **Recommendations AI** for personalization
- **Natural Language API** for content classification

Scalable Infrastructure

- **GKE Autopilot** for managed container orchestration
- **Cloud CDN** for global content delivery
- **Cloud Load Balancing** for high availability
- **Cloud Armor** for DDoS protection

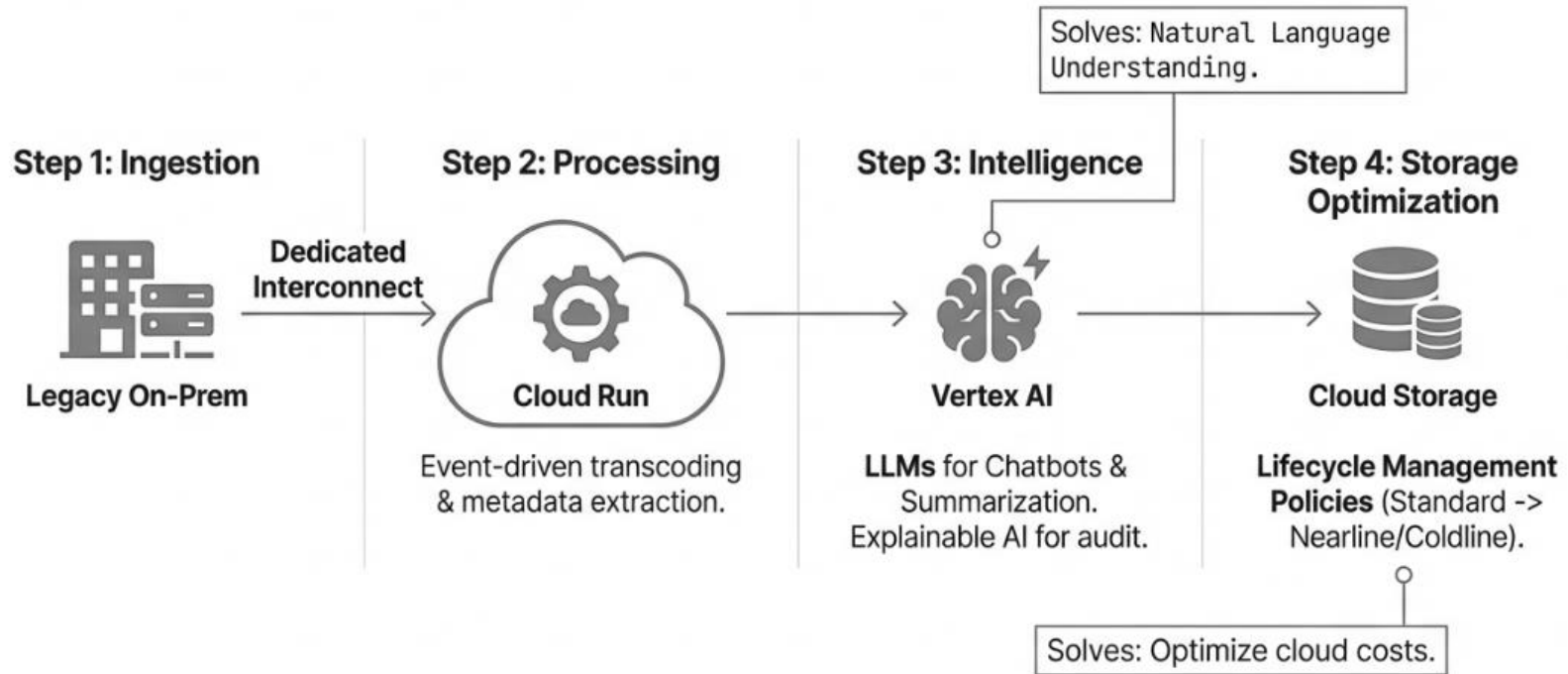
Analytics & Insights

- **BigQuery** as central data warehouse
- **Looker** for business intelligence dashboards
- **Dataflow** for real-time streaming analytics
- **Cloud Logging** for operational insights

Exam Focus Areas

Expect questions on: content processing pipelines, Vertex AI integration, Cloud Storage lifecycle management, GKE scaling patterns, and BigQuery analytics for media consumption patterns.

Altostrat GenAI Content Pipeline



Case Study: Cymbal Retail

Company Profile

Growing online retailer with large product assortment spanning multiple retail sub-verticals. Managing extensive product catalog is a constant challenge.

Core Challenges

- Manual catalog management is error-prone
- Data silos limit customer journey visibility
- Difficult to integrate new technologies
- IVR system handles initial customer calls
- Call center agents manually enter orders

Solution Concept (3 Pillars)

- **Catalog Enrichment:** Gen AI to auto-generate product attributes, descriptions, and images from supplier data
- **Conversational Commerce:** AI virtual agents for personalized shopping via natural language
- **Stack Modernization:** Cloud infrastructure, secure data handling, 3rd-party integrations

Key Constraints

- Secure handling of sensitive customer data
- Scale efficiently during peak seasons
- Minimize downtime during integration
- GDPR and privacy compliance

Case Study: Cymbal Retail

Current Infrastructure

- Mix of on-premises and cloud-based systems
- IVR system for initial customer calls
- Call center for manual order entry
- Multiple product databases (siloes)
- Traditional e-commerce platform

Monitoring Tools

- Grafana for visualization
- Nagios for infrastructure monitoring
- Elastic Stack for log management
- Various open-source tools (fragmented)

Current State Pain Points

- Manual processes are time-consuming and error-prone
- Data silos prevent unified customer journey view
- Integrating new technologies is difficult
- Inconsistent product information across channels

Technical Debt

- Legacy IVR lacks modern AI capabilities
- No centralized data platform
- Manual catalog updates from supplier feeds
- Limited personalization capabilities

Case Study: Cymbal Retail

Catalog Enrichment with Gen AI

- **Vertex AI** with Gemini for product description generation
- **Vision AI** for image analysis and tagging
- **Document AI** for supplier data extraction
- **Cloud Storage** for product media assets

Conversational Commerce

- **Dialogflow CX** for conversational AI agents
- **Discovery AI** for product search and retrieval
- **Recommendations AI** for personalized suggestions
- **CCAI** (Contact Center AI) for call center

Data Platform

- **BigQuery** for unified customer data warehouse
- **Dataflow** for real-time data pipelines
- **Pub/Sub** for event-driven architecture
- **Data Catalog** for metadata management

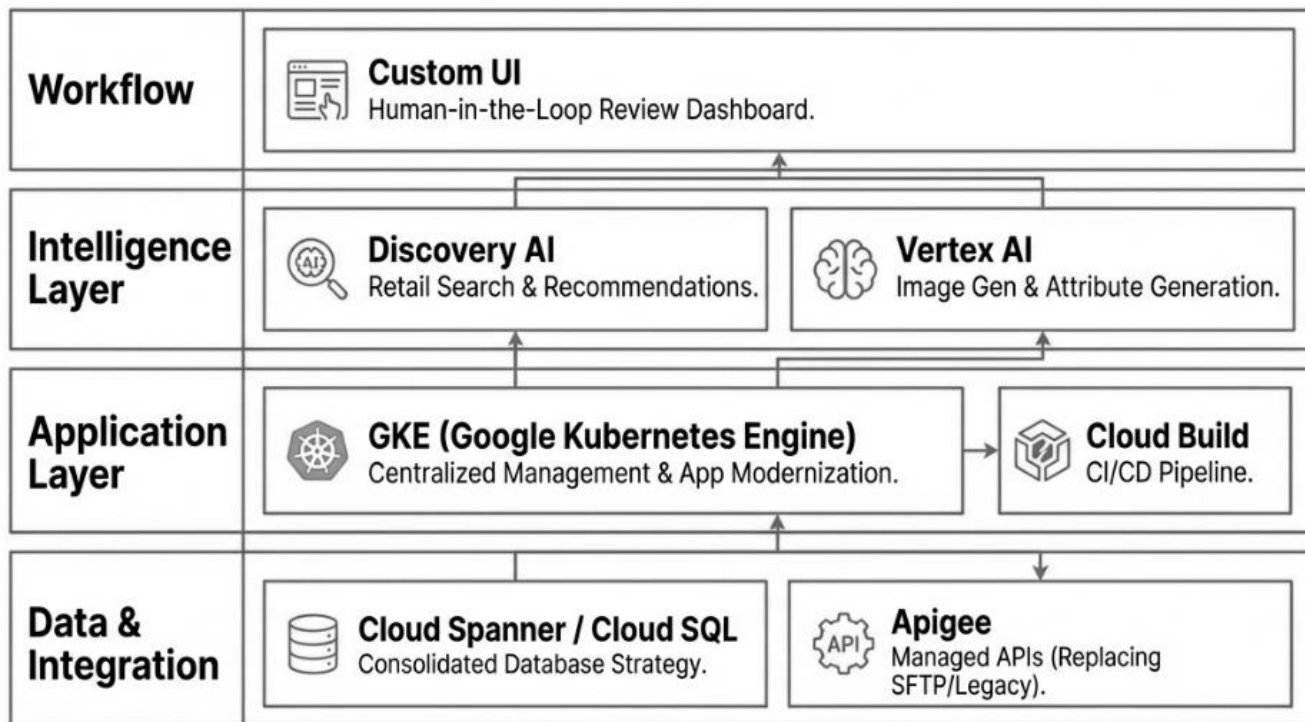
Security & Compliance

- **Sensitive Data Protection** for PII handling
- **Secret Manager** for credentials
- **Cloud Armor** for web application protection
- **VPC-SC** for data exfiltration prevention

Exam Focus Areas

Expect questions on: Gen AI for catalog automation, Dialogflow CX design, Discovery AI configuration, PII protection with Sensitive Data Protection, and scaling for peak retail seasons.

Cymbal Retail Commerce Stack



Case Study: EHR Healthcare

Company Profile HIPAA

Leading provider of electronic health record (EHR) software serving multi-national medical offices, hospitals, and insurance providers via SaaS model.

Business Drivers

- Rapid growth requiring scalable infrastructure
- Data center lease expiring (migration trigger)
- Need for improved disaster recovery
- Faster software deployment cycles
- Onboard new insurance providers quickly

Solution Concept

- Migrate from colocation to Google Cloud
- Implement infrastructure-as-code automation
- Establish CI/CD for faster deployments
- Unified monitoring and alerting
- Maintain HIPAA compliance throughout

Business Requirements

- 99.9% availability for customer-facing systems
- Centralized visibility and proactive monitoring
- Reduced latency for all customers
- Healthcare trend insights and reporting
- Maintain regulatory compliance

Case Study: EHR Healthcare

Current Infrastructure

- Multiple colocation facilities (one lease expiring)
- Containerized web apps on Kubernetes clusters
- Mixed database environment:

MySQL MS SQL Server Redis MongoDB

Identity & Monitoring

- Microsoft Active Directory for user management
- Various open-source monitoring tools
- Email alerts (often ignored)
- No centralized observability platform

Legacy Systems (Not Migrating)

- File-based integrations with insurance providers
- API-based integrations (on-premises)
- Scheduled for replacement over several years
- Requires hybrid connectivity to cloud

Technical Requirements

- Maintain legacy insurance provider connectivity
- Consistent container-based app management
- Secure, high-performance hybrid connectivity
- Unified logging, monitoring, and alerting
- Dynamic scaling and provisioning

Case Study: EHR Healthcare

Compute & Containers

- **GKE** with auto-scaling for containerized EHR apps
- **Artifact Registry** for container images
- **Cloud Build** for CI/CD pipelines
- **Cloud Deploy** for managed deployments

Hybrid Connectivity

- **Dedicated Interconnect** (2 connections, 2 metros)
- **Cloud VPN** as backup connectivity
- **Cloud Router** for dynamic routing
- **Private Google Access** for GCP APIs

Data & Identity

- **Cloud SQL** for MySQL/PostgreSQL migration
- **Memorystore** for Redis migration
- **Google Cloud Directory Sync** with AD
- **BigQuery** for healthcare analytics

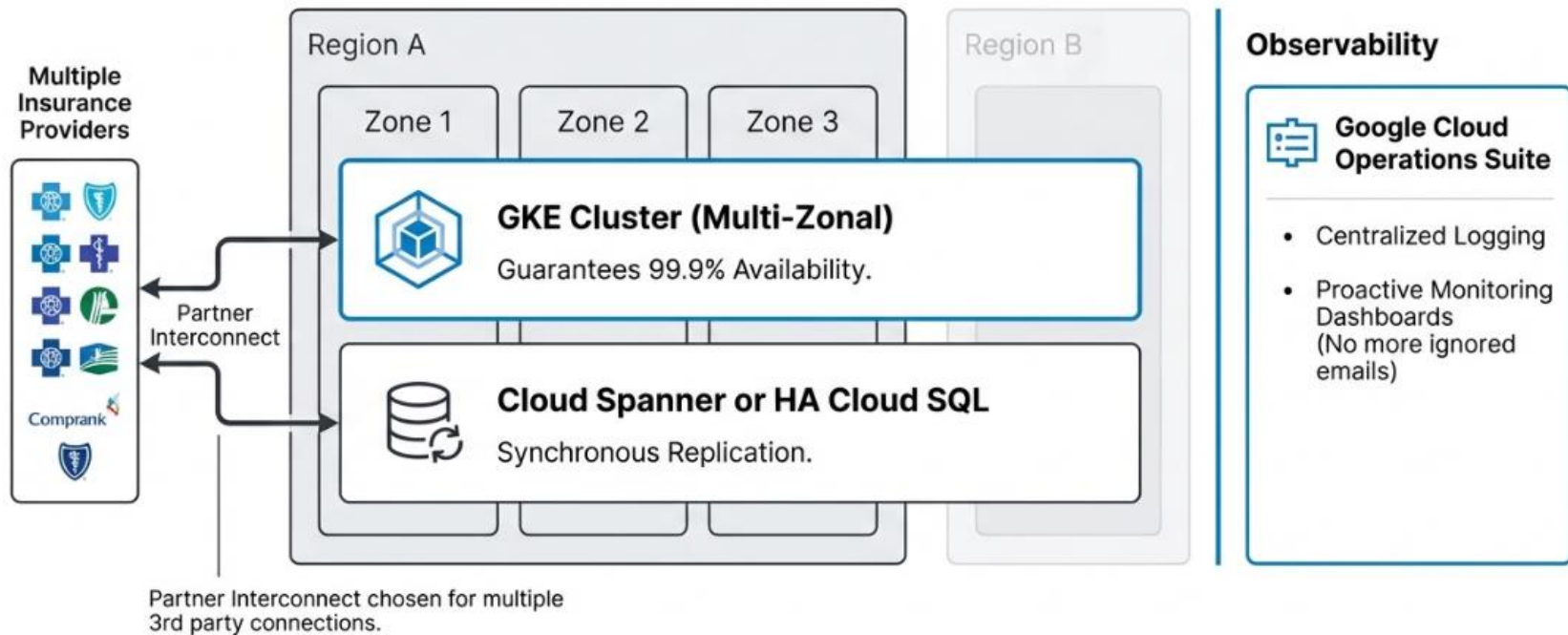
Operations & Compliance

- **Cloud Monitoring** for unified observability
- **Cloud Logging** with log-based metrics
- **BAA** (Business Associate Agreement) for HIPAA
- **Apigee** for insurance provider API management

Exam Focus Areas

Expect questions on: Dual Dedicated Interconnect for HA, GKE migration patterns, HIPAA compliance (BAA, covered services), Cloud Directory Sync with AD, and disaster recovery strategies.

EHR Healthcare HA Design



Case Study: KnightMotives Automotive

Company Profile IoT

Automotive company providing connected-car services. Gathers telemetry from vehicles (location, sensor readings) to improve maintenance and driver services.

Business Drivers

- Millions of telemetry data points per minute
- Real-time vehicle health monitoring
- Predictive maintenance insights
- Partner data sharing (insurers, dealerships)
- Fleet management dashboards

Solution Concept

- High-volume telemetry ingestion at scale
- Real-time lookups for vehicle status
- Analytics on historical data patterns
- Secure API sharing with partners
- Edge computing for in-vehicle processing

Key Constraints

- Handle intermittent vehicle connectivity
- Low-latency real-time queries
- Cost-effective long-term storage
- Secure data sharing with external partners

Case Study: KnightMotives Automotive

Vehicles (Telemetry)



Pub/Sub (Ingestion)



Dataflow (Processing)



Bigtable (Real-time)



BigQuery (Analytics)

Data Ingestion Layer

- **Pub/Sub** for horizontally scalable ingestion
- Handles intermittent connectivity (buffer/retry)
- Dead-letter topics for failed messages
- IoT Core for device management (optional)

Real-Time Processing

- **Dataflow** for streaming ETL
- Filtering, aggregating, enriching telemetry
- Windowed computations for alerts
- Auto-scaling based on data volume

Storage Strategy (Key Exam Topic)

- **Bigtable** for time-series data (keyed by vehicle ID + timestamp) - high-speed writes, per-vehicle queries
- **BigQuery** for large-scale analytics on aggregated data - fleet-wide reports, trend analysis
- Recent data in Bigtable for quick access; older data periodically exported/summarized to BigQuery for cost control

Case Study: KnightMotives Automotive

Telemetry Ingestion

- **Pub/Sub** for scalable message ingestion
- **Dataflow** streaming for real-time processing
- **Cloud Functions** for event-driven alerts
- **Pub/Sub Lite** for cost-optimized zones

Time-Series Storage

- **Bigtable** keyed by vehicleID + timestamp
- **BigQuery** for historical analytics
- **Cloud Storage** for raw data archival
- Lifecycle policies for cold data tiering

AI/ML Capabilities

- **Vertex AI** for predictive maintenance models
- **BigQuery ML** for anomaly detection
- **AutoML Tables** for driver behavior analysis
- Real-time inference for vehicle health alerts

Partner API Access

- **Apigee** for managed API gateway
- Rate limiting and quota management
- OAuth 2.0 for partner authentication
- Analytics Hub for secure data sharing

Exam Focus Areas

Expect questions on: Bigtable vs BigQuery for time-series (real-time vs analytics), Pub/Sub scaling patterns, Dataflow streaming pipelines, Apigee for partner APIs, and edge computing architectures.

Case Studies Cheat Sheet

Altostrat Media

Industry: Digital **media, streaming**

Key Services: GKE, Vertex AI, BigQuery, Cloud Storage, Cloud CDN

Challenges: AI integration, personalization, storage optimization

Exam Focus: Content pipelines, Vertex AI, recommendations

Cymbal Retail

Industry: E-commerce, **retail**

Key Services: Dialogflow CX, **Discovery AI, Vertex AI**, DLP

Challenges: Catalog automation, conversational commerce, PII

Exam Focus: **Gen AI** catalogs, CCAI, privacy, seasonal scaling

EHR Healthcare

Industry: Healthcare SaaS, medical records

Key Services: GKE, Dedicated **Interconnect**, Cloud SQL, Apigee

Challenges: **HIPAA** compliance, **hybrid** connectivity, DR, CI/CD

Exam Focus: Dual Interconnect HA, BAA, Directory Sync

KnightMotives Automotive

Industry: Automotive **IoT**, connected cars

Key Services: Bigtable, **Pub/Sub, Dataflow**, BigQuery, Apigee

Challenges: **High-volume telemetry**, real-time queries, partner APIs

Exam Focus: Bigtable vs BigQuery, streaming, **time-series**

Case Studies Cheat Sheet

Company	Compute Strategy	Database Shift	AI Use Case	Key Constraint
Altostrat	Cloud Run + GKE	BigQuery (Analytics)	Creative (Video/Summarization)	Cost & Storage
Cymbal	GKE (App Mod)	Spanner (Consolidation)	Structured (Catalog/Search)	Sprawl & Toil
EHR	Multi-Zonal GKE	SQL/NoSQL (Scale/HA)	Predictive (Trends)	Availability (99.9%)
KnightMotives	Hybrid / Anthos	BigQuery (Monetization)	Autonomous (Vision)	Latency & Legacy

Segment 9: Wrap-up



Objectives

- Wrap-up and study resources
- Exam tips and next steps

Study Resources

- cloudskillsboost.google: Hands-on labs
 - Google Cloud documentation pages (ignore Preview stuff) 
 - Gemini/Claude/ChatGPT (proceed with caution)
-
- Getting Started with Google Cloud Platform (LiveLessons)
 - Associate Cloud Engineer Crash Course (LiveLessons)